



Capable People, Capable Communities

Micro-Credentials in Software Development

Certificate in Software Development

MCSD51

Project Report

EasyOrdering

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EXECUTIVE SUMMARY

In this project, I built a web-based ordering system for a restaurant, making it easier for customers to place their orders. I used basic HTML, CSS, Bootstrap, MySQL, Node.JS and JavaScript to develop this app. This is a self-sponsored, individual project developed in an iterative manner. I created the Use Cases based on my personal knowledge and experience related to restaurants.

INTRODUCTION

Many restaurants in New Zealand still rely on traditional methods of customer service, such as printed menus and manual order taking. While this approach is familiar, it often results in inefficiencies, miscommunication, and increased labour costs. In contrast, with the rapid advancement of digital technologies, many countries — especially in Asia — have widely adopted electronic menus and self-service ordering systems to enhance both customer experience and operational efficiency.

A notable example is China, where WeChat Mini Programs have revolutionised the way customers interact with restaurants. Customers simply scan a QR code at the table using their smartphones, which opens a digital menu within the app. They can browse the menu, place orders, make payments, and even request service — all from their device. This end-to-end digital workflow significantly reduces the reliance on waitstaff and streamlines restaurant operations. Implementing such a system offers several advantages:

- Cost Efficiency
 - ✓ Eliminates the need for printed menus and frequent reprinting due to menu changes.
 - ✓ Reduces labor costs by minimizing or even eliminating the need for waitstaff during peak hours.
- Real-Time Flexibility
 - ✓ Allows restaurant owners to update menu items, prices, or promotions instantly.
 - ✓ Seasonal items or limited-time offers can be added without delays or printing costs.
- Improved Customer Experience
 - ✓ Enables customers to place orders at their own pace, without pressure or errors in communication.
 - ✓ Ideal for customers who prefer a contactless or quieter dining experience.
- Operational Efficiency
 - ✓ Orders are submitted directly to a receptionist dashboard, reducing human error.
 - ✓ Featured or recommended dishes can be highlighted to promote high-margin items or chef's specials.

The goal of this project is to develop a web-based restaurant ordering system tailored for local New Zealand businesses. Customers will scan a QR code at their table, which directs them to a mobile-friendly web application. Through the interface, they can view a dynamic and categorised digital menu, place and customise their orders, see highlighted or promotional dishes and submit their order directly to the restaurant's staff or kitchen system.

The system will also provide restaurant staff with a management interface to handle incoming orders, update menu items, and monitor order statuses. This solution aims to improve customer satisfaction through convenience and clarity, while also helping restaurant owners to cut costs, optimize staff allocation, modernize their operations and stay competitive in an increasingly digital landscape

INITIAL PROPOSAL

1. INTRODUCTION

Many restaurants in New Zealand continue to use traditional methods for serving customers. However, with the rapid development of technology, electronic menus and ordering systems have become increasingly popular in many countries. For instance, almost all restaurants in China now use WeChat mini programs for ordering. Customers simply scan a QR code with WeChat, which opens a digital menu within the mini program. From there, they can browse the menu, place orders, and make payments — essentially handling everything except the cooking.

This system offers several advantages. First, it eliminates the need to print physical menus. Second, it allows restaurants to easily update their offerings, such as adding or removing dishes or applying discounts, all without incurring additional costs. Third, it reduces the need for staff, as fewer waiters — or even none — may be required. Finally, it caters to customers who prefer a quieter dining experience without waiter interactions, giving them the freedom to order at their own pace.

The goal of this proposal is to develop a web-based ordering system for a restaurant. Customers will be able to scan a QR code at their table, access the website, view the menu, and place their orders. The system will also highlight featured dishes and automatically notify the receptionist of new orders.

This solution aims to enhance the customer experience while helping restaurant owners reduce operational costs and improve overall efficiency.

2. SCOPE

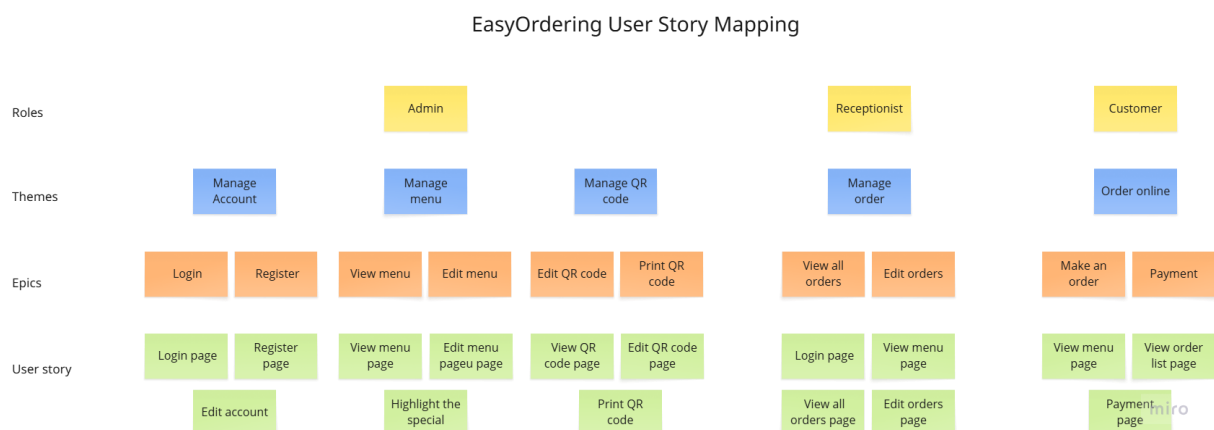
The completed project will have a home page providing some information about what the webpage is about & how to navigate it and show the top 10 ordered dishes. Everyone can visit the home page. There is a login link on the home page. Admin and receptionist users can log in through the link and log in. If the Admin has logged in, he/she could see the Menu, QR code and Account links. That means there are three pages that Admin can edit. If a Receptionist has logged in, he/she could see the Menu(different from Admin) and Orders page. The Receptionist can place an order for a table of customers on the Menu page and view all orders on the Orders page. Customers don't have to log in. After scanning a code and going into the Menu page where they can place an order and all they will see order list on the same page. After a customer places an order, the Receptionist will receive a notification on the Orders page. There is a payment page where Customers can pay online, but because I do not have a payment gateway, I will just finish the page without functionality.

In total, there are at least nine pages(Home, Login, Register, Editing Menu, Account, QR code, Orders, Menu, Payment).

The completed project will also have a database with at least five tables, one for accounts, one for orders, one for code, and two for menu.

3. USER STORIES AND USE CASES

According to the requirement, I can map them into different levels, like the following User Story Mapping.



As an admin, I want to log in to the system so I can manage the accounts, menu and QR codes.

As an admin, I want to be able to create some accounts for my employees so I can view menus and orders.

As an admin, I want to be able to delete some accounts when the employee leaves my restaurant so it is safe.

As an admin, I want to be able to edit the menu so I can add, delete, pause or modify some dishes. For example, I can give a special price for a chocolate cake today.

As an admin, I want to be able to view the special dishes so my customers can see them more easily.

As an admin, I want to be able to view the menu that my customer can see so I can know the effect.

As an admin, I want to be able to create a QR code so I can have a new QR code when I have a new table.

As an admin, I want to be able to view all QR codes so I can see that all tables have related QR codes.

As an admin, I want to be able to print a QR code so I can paste it on the table that the customers can scan and into our website.

As a receptionist, I want to log in to the system so I can view the menu and all orders.

As a receptionist, I want to view the menu so I can order dishes for customers.

As a receptionist, I want to be able to view all orders, so I can tell the chef to cook them.

As a receptionist, I want to be able to edit an order, so I can update the status of dishes, the payment status of the order, and delete or add some dishes according to the customer's needs.

As a customer, I want to be able to view the menu, so I can choose what dishes I want.

As a customer, I want to be able to place an order, so the restaurant knows that I have finished the order.

As a customer, I want to be able to add some dishes to the order, so I can have more dishes.

As a customer, I want to be able to pay my bill online, so I don't need to pay at the receptionist.

4. ETHICAL AND CULTURAL IMPACT

There are five common ethical issues in software development: addictive design, corporate ownership of personal data, algorithmic bias, weak cybersecurity and protection of personally identifiable information (PII), and overemphasis on features⁽²⁾.

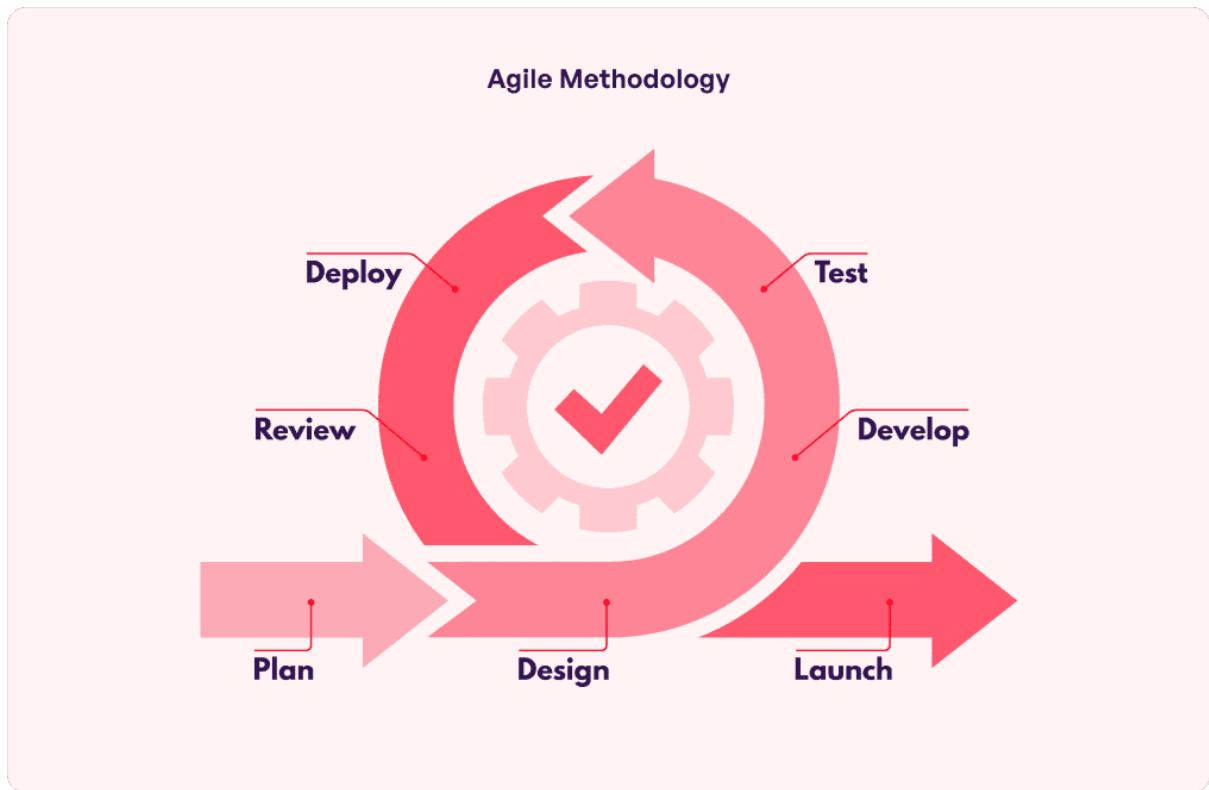
Regarding the website I am developing, it avoids several of these concerns. Since users are not required to log in to place orders and the payment page is only used to update the order status (without processing actual transactions), the system does not collect or store any personal data. Therefore, issues related to corporate ownership of personal data and weak cybersecurity/PII protection are not applicable.

Additionally, because the system does not track or profile users in any way, algorithmic bias is not a concern. The website simply serves as a tool for customers who have already visited the restaurant to place and manage orders—it does not attempt to influence behavior or encourage repeated engagement. As such, addictive design is also not relevant.

Finally, in terms of functionality, the system is intentionally kept minimal, providing only essential features necessary for its core purpose. Since it does not include unnecessary or excessive features, the issue of overemphasis on features does not apply at this stage.

5. SDLC

I decided to use the Agile software development methodology because it has been designed with speed, flexibility, rapid prototyping, collaboration, and adaptability in mind – the keys to any startup success. Moreover, agile SDLC ensures that the end product is customizable to suit the changing needs. This is perfect since they are constantly updating, and evolving. Also, A big project can be broken into smaller 'sprints' or time-boxed cycles. Developers can prioritise any feature of software initially or at any given point rather than following a set bureaucratic chain⁽¹⁾. Finally, Agile methodology is the most popular model now so I would like to try.



6. PROJECT TIMELINE

Sprint 1: It should finish the basic skeleton. This means some webpages are set up, the login page, the account page and the database are set up. The Admin user can log in and add/delete/view all accounts. The Receptionist user can log in.

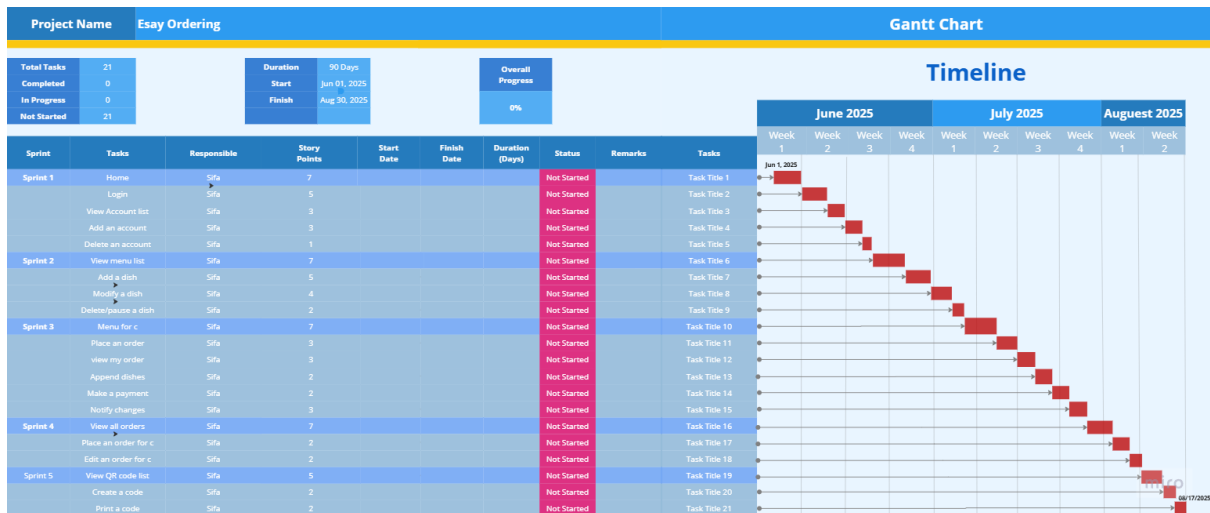
Sprint 2: It should finish the pages related to the menu management. This means the Admin user can log in and add/delete/view the menu and set some special functions for the menu. The Receptionist user can log in and view the menu.

Sprint 3: It should finish the pages related to customers making an order. This means a customer can view the menu, place an order, append orders, and make a payment. It should notify the reception automatically that there are new orders.

Sprint 4: It should finish the pages related to the receptionist managing orders. This means the receptionist user can view all orders and get messages when changes happen. He can also take an order for customers, modify the orders and manage the status when people pay by cash.

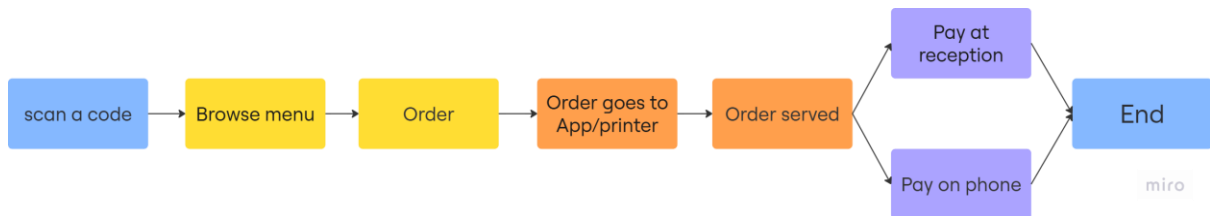
Sprint 5: It should finish the pages related to QR code management. This means the Admin user can view all codes and create a new code. Anyone can scan the code to the website.

The following Gantt Chart shows the detailed timeline.



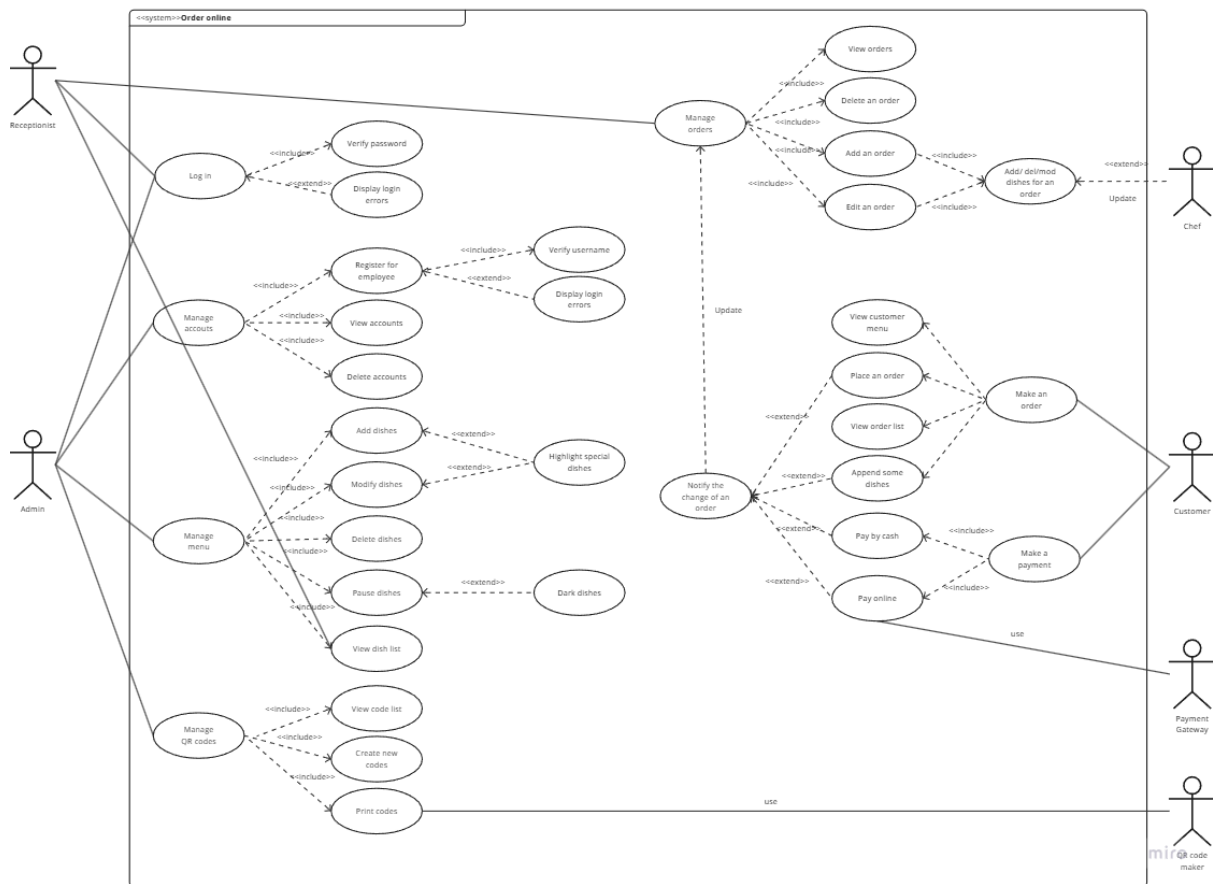
7. DIAGRAMS

The following Flow chart shows the most important flow of the system.



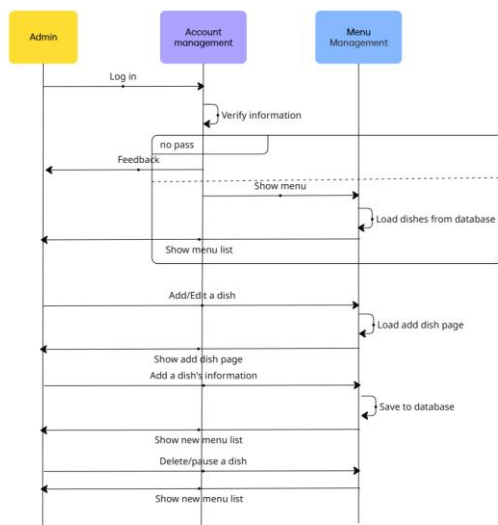
The following Use Case chart shows the main use cases.

EasyOrdering Use Case

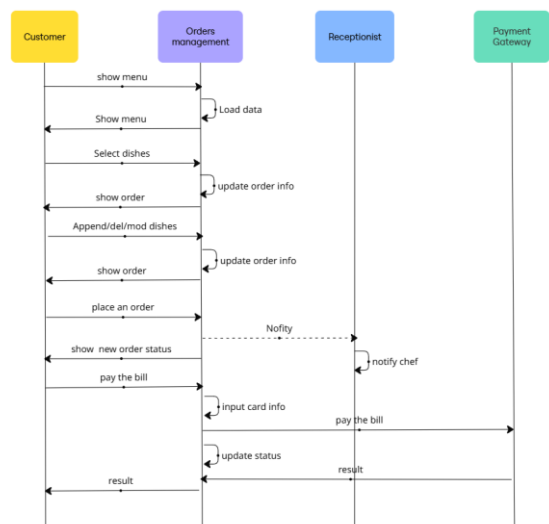


The following Sequence chart shows the primary sequences.

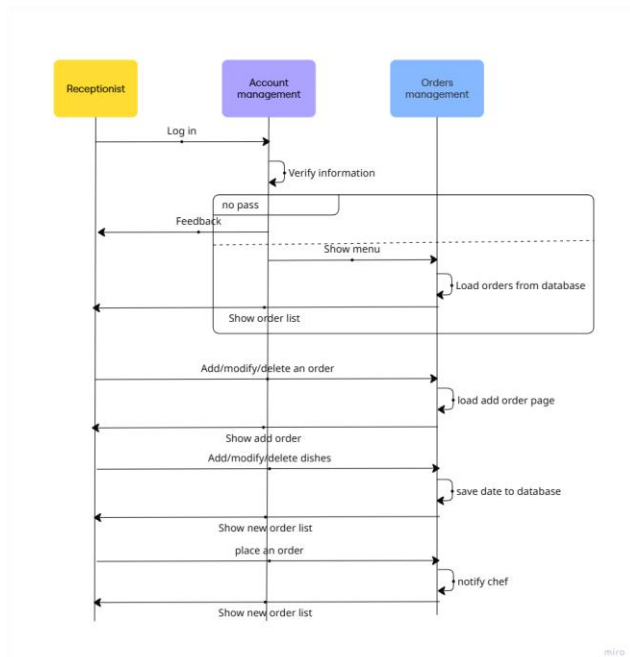
Admin Edit menu



Customer place a order



Receptionist schedule orders



8. REFERENCES

(1) Agile methodology. Published April 15, 2024. Retrieved from:

<https://www.simublade.com/blogs/sdlc-methodologies/>

(2)Retrieved from:

[5 examples of ethical issues in software development | TechTarget](#)

VARIATIONS IN USER REQUIREMENTS FROM THE INITIAL PROPOSAL

I've reorganised the table after adding and refining some user stories.

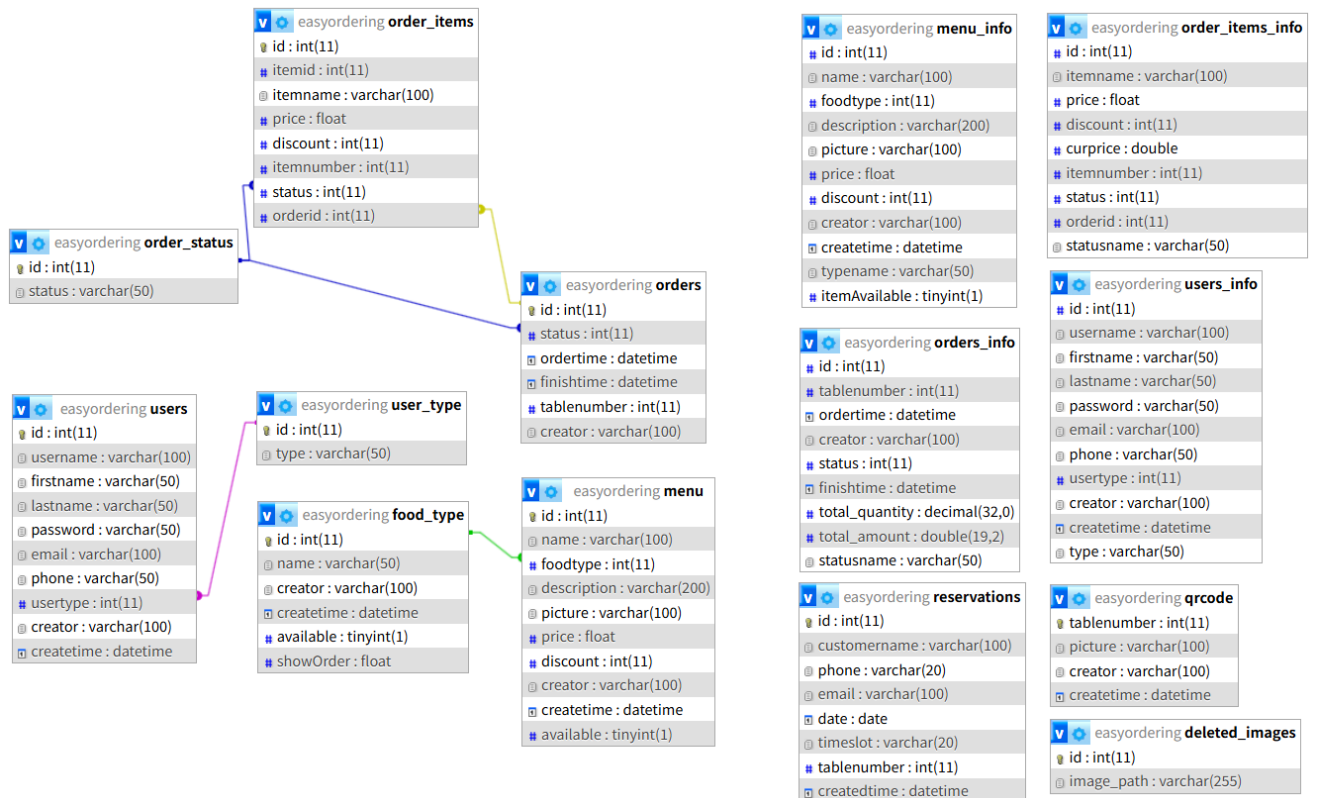
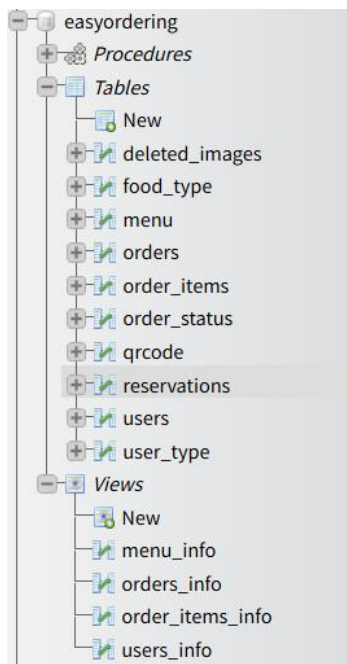
Role	Action	Benefit
All users	I want to visit the website and see the home page	I can know basic information and the top 5 popular items.
All users	I want to know information about the restaurant.	I can contact or go to the restaurant.
Admin and Receptionist	I want to log in to the system	I can do some job with my role.
Admin and Receptionist	I want to get back my password when I forget my password	I can retrieve my account
Admin and Receptionist	I want to change my account information	I can change my password for safety and update my profile.
Admin	I want to view all current accounts so I can delete them and I can create a new account.	I can manage accounts according to the current employer.
Admin	I want to have error tips when I create a new account, like two passwords must be the same	I can create the right account.

Admin	I want to view the current menu and special information	I can know how customers see the menu
Admin	I want to add, delete, list, unlist and change the show order of a food type	I can adjust the menu dynamically according to the season and the chef
Admin	I want to add, delete, list, unlist and edit the menu to the current food type.	I can adjust the menu dynamically according to the season and the chef
Admin	I want to have error messages when I create a new dish and tips when I delete, list or unlist a dish.	I can create the right dish and avoid wrong actions.
Admin	I want to view, add, delete, and print a QR code.	I can paste it on the table that the customers can scan and into our website.
Guest	I want to view the menu	I can know what dishes I want to order
Customer	I want to scan a QR code to enter the system	I can view the menu
Customer	We want to view today's menu, including price and discount. We can add items to our cart	We can select items for our dinner on our phones, like everyone has a menu.
Customer	I want to know how many items and the total price are when I add the items	I can know how much I have selected
Customer	I want to view my cart and delete items from my cart	I can decide what items I would like to order
Customer	I want to place an order and know the status of every dish	The restaurant can cook my dishes
Customer	I want to pay online	I needn't to pay at the counter.
Receptionist	I want to select dishes for customers	I can choose items for customers
Receptionist	I want to know how many items and the total price are when customers add the items	I can tell the customers the number and amount
Receptionist	I want to view my cart and delete items for customers	I can confirm with customers
Receptionist	I want to place an order and know the status of every dish	I can tell customers the status of the dishes
Receptionist	I want to search the history's orders	I can check the history's information
Receptionist	I want to have a voice notify me and a tip when new orders have been placed.	I can know new orders immediately
Receptionist	I want to view today's orders and receive orders	I can tell that orders have been received by me
Receptionist	I want to print new orders	I can tell the chef to cook them
Receptionist	I want to delete items for customers	I can provide services for customers who would like to place an order with me
Receptionist	I want to delete or cancel orders	I can know the new status when customers change their ideas
Receptionist	I want to change the status to ready when a dish has been served.	I can record which items have been served.
Receptionist	I want to finish an order when customers pay at the counter	I can know which orders have finished
Receptionist	I want to print new items	I can give the paper to the chef
Receptionist	I want to print a receipt	I can give it to the customer
Receptionist, Guest and Customer	I want to make a reservation	I can book a table in advance
Receptionist	I want to cancel a reservation	I can cancel reservations for customers

DESIGN DOCUMENTS

Database

The system contains 10 tables and 4 views. Their relationships are shown below.



There are also 6 Triggers used for deleting food type, menu, QR code and updating status.

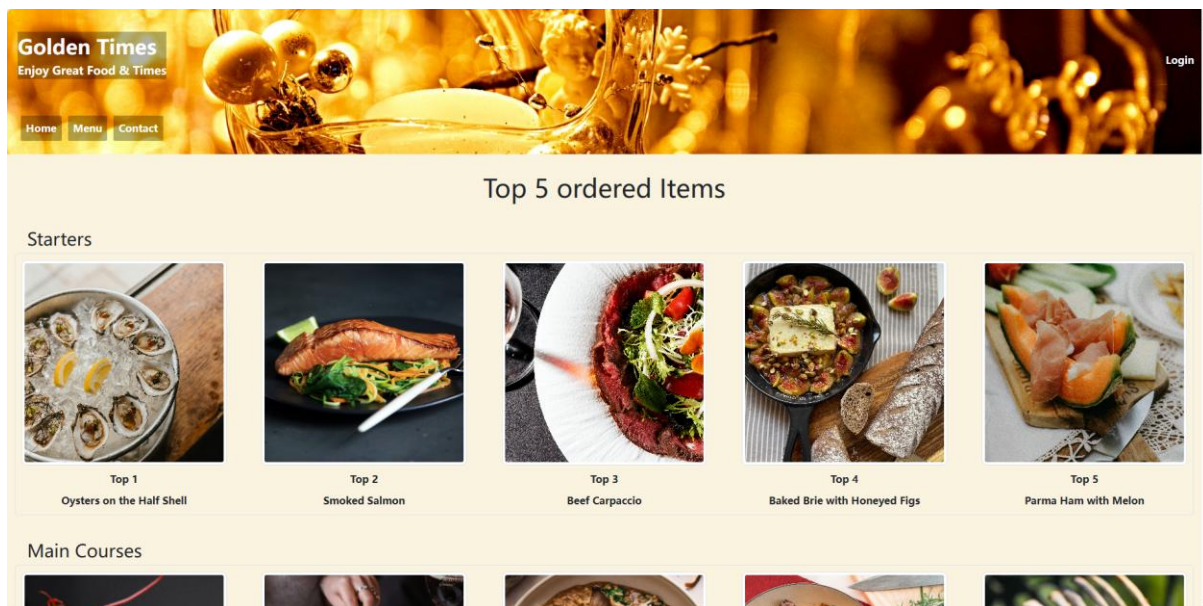
	Name	Table	Time	Event	
☐	before_delete_foodtype	food_type	BEFORE	DELETE	Edit Export Drop
☐	before_delete_menu	menu	BEFORE	DELETE	Edit Export Drop
☐	before_delete_qrcode	qrcode	BEFORE	DELETE	Edit Export Drop
☐	before_order_delete	orders	BEFORE	DELETE	Edit Export Drop
☐	trg_order_items_status_update	order_items	BEFORE	UPDATE	Edit Export Drop
☐	update_order_items_status	orders	AFTER	UPDATE	Edit Export Drop

A stored procedure used for counting the top n items for each food type and show on the Home page.

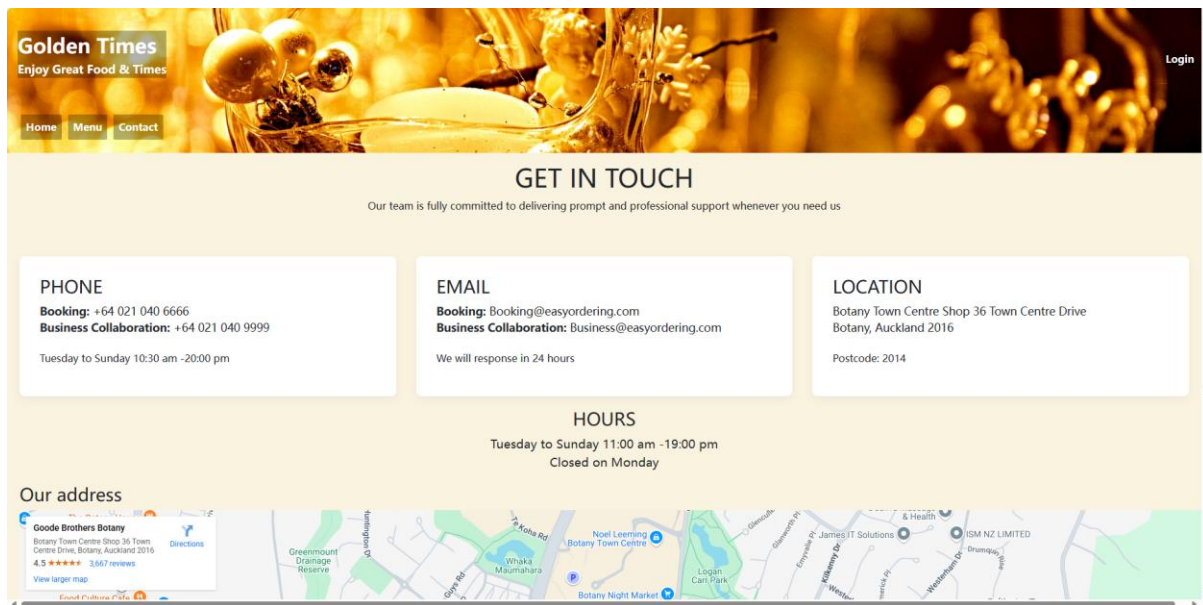
	Name	Type	Returns	
☐	SelectTopItemsByType	PROCEDURE		Edit Execute Export Drop

Screen shots

All users, including Guest, Customer, Admin and Receptionist, can see the Home page. This page shows the top 5 popular items in each food type. These items are counted from finished orders in the database.

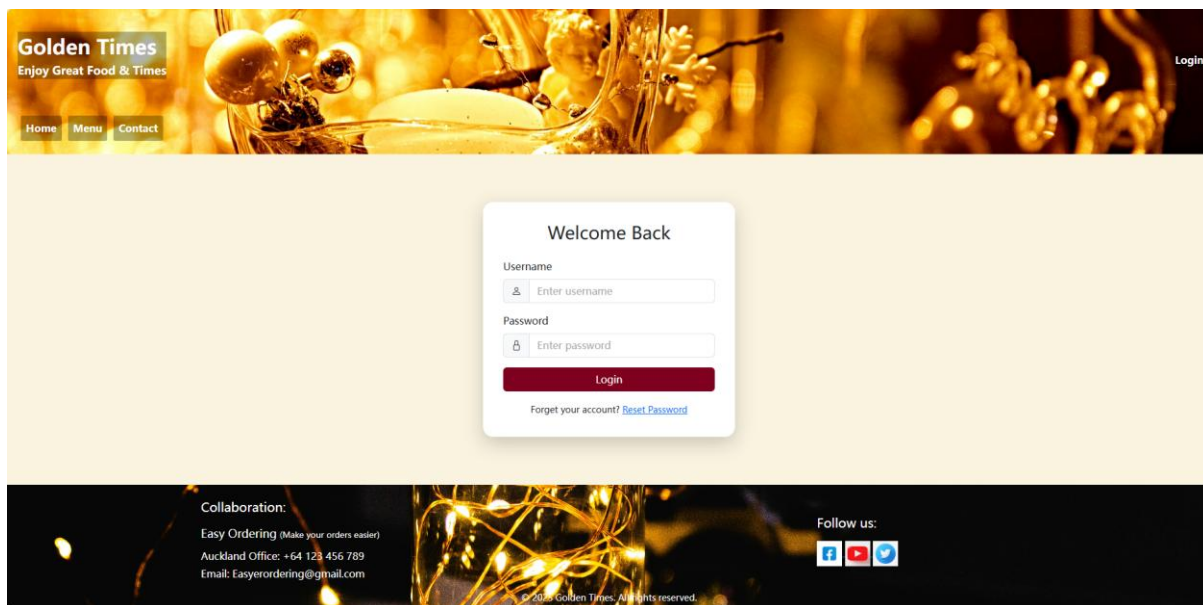


All users can see the Contact page. This page shows the basic information about the restaurant.



The screenshot shows the 'Contact' page of the Golden Times website. The header features the restaurant's name 'Golden Times' with the tagline 'Enjoy Great Food & Times', a 'Login' button, and navigation links for 'Home', 'Menu', and 'Contact'. The main section is titled 'GET IN TOUCH' with a commitment statement: 'Our team is fully committed to delivering prompt and professional support whenever you need us'. Below this, three white boxes provide contact details: 'PHONE' (Booking: +64 021 040 6666, Business Collaboration: +64 021 040 9999, Tuesday to Sunday 10:30 am -20:00 pm), 'EMAIL' (Booking: Booking@easyordering.com, Business Collaboration: Business@easyordering.com, We will response in 24 hours), and 'LOCATION' (Botany Town Centre Shop 36 Town Centre Drive, Botany, Auckland 2016, Postcode: 2014). A 'HOURS' section states 'Tuesday to Sunday 11:00 am -19:00 pm' and 'Closed on Monday'. The 'Our address' section includes a Google Maps snippet for 'Goode Brothers Botany' and a larger map showing the location at 'Botany Town Centre'.

This is the login page for Admin and Receptionist. They have to use a Username and Password to log in.



The screenshot shows the login page of the Golden Times website. The header is identical to the contact page. The main content area features a 'Welcome Back' message above a login form. The form has fields for 'Username' (with a hint 'Enter username') and 'Password' (with a hint 'Enter password'), a red 'Login' button, and a link for 'Forgot your account? Reset Password'. The footer contains collaboration information: 'Easy Ordering (Make your orders easier)', 'Auckland Office: +64 123 456 789', 'Email: Easyordering@gmail.com', and social media links for Facebook, YouTube, and Twitter. A copyright notice '© 2018 Golden Times. All rights reserved.' is also present.

If users forget their Username or Password, they can click the Reset Password link to get their Username and Password. On the Reset Page, they should input their registered email to retrieve their accounts.

Golden Times
Enjoy Great Food & Times

Home Menu Contact

Login

Please input email

Email

sifazhang.nz@gmail.com

Reset

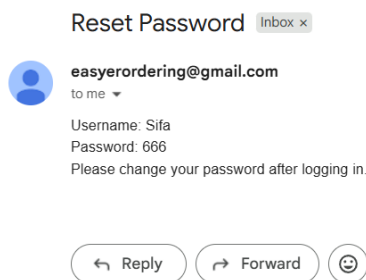
Collaboration:
Easy Ordering (Make your orders easier)
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Email: Easyordering@gmail.com

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Then they will get an email like this:



If a user logs in as an Administrator, he/she will see his/her Username and the role on the top right page. The Administrator roles can see Home, Menu, Accounts, QRcode, My account and Contact pages. On the Accounts page, they can see all accounts in the system. They can delete an existing user.

Golden Times
Enjoy Great Food & Times

Home Menu Accounts QRcode My account Contact

Welcome Administrator: Admin Logout

All Accounts' Information

User name	First name	Last name	Phone	Email	User type	Creator	Create time	Action
Admin	Sifa	Zhang	0210408866	easyordering@gmail.com	Administrator	Admin	12/06/2025, 11:54:00 am	Delete
Serena	Serena	Jia	021555666	serena@gmail.com	Administrator	Admin	23/06/2025, 7:31:43 am	Delete
Sifa	Sifa	Zhang	021888888	Sifazhang.nz@gmail.com	Receptionist	Admin	12/06/2025, 3:22:11 am	Delete

Add a user

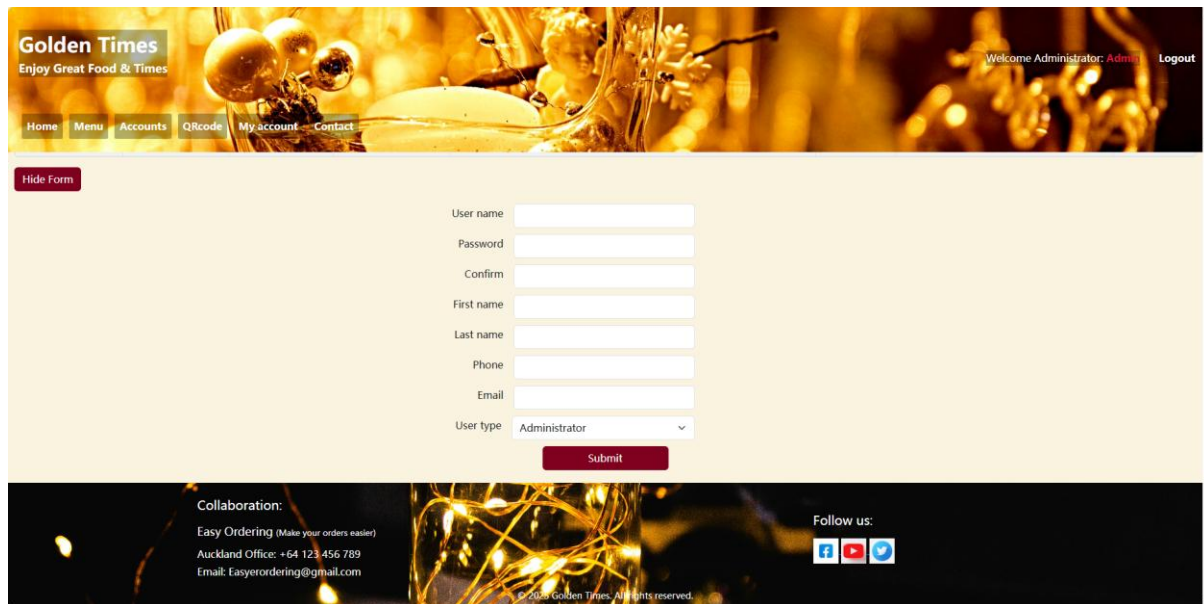
Collaboration:
Easy Ordering (Make your orders easier)
Auckland Office: +64 123 456 789
Email: Easyordering@gmail.com

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They can create a new user. This page will check that all input boxes have been filled in and that the same content on the Password and the Confirm input boxes. Otherwise, the system will tip messages. All receptionists can update their account information like Administrators.



Golden Times
Enjoy Great Food & Times

Welcome Administrator: Admin Logout

Home Menu Accounts QRcode My account Contact

Hide Form

User name
Password
Confirm
First name
Last name
Phone
Email
User type Administrator

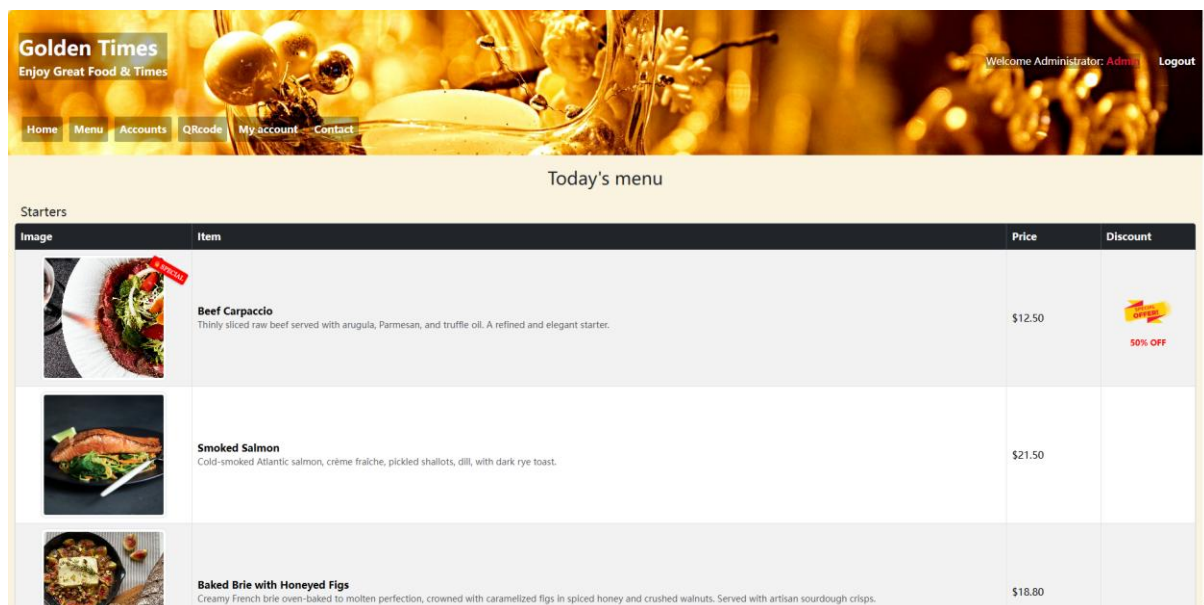
Submit

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The administrators can see Today's menu by clicking Menu-Overview. They will see all the listed food types and items, including price and discount, but they can not add items to the cart. If an item has a discount, a Special picture will show on the top right of its image and in the Discount column. All the items' images can be clicked to zoom in. The guests can see the same page by clicking Menu.



Golden Times
Enjoy Great Food & Times

Welcome Administrator: Admin Logout

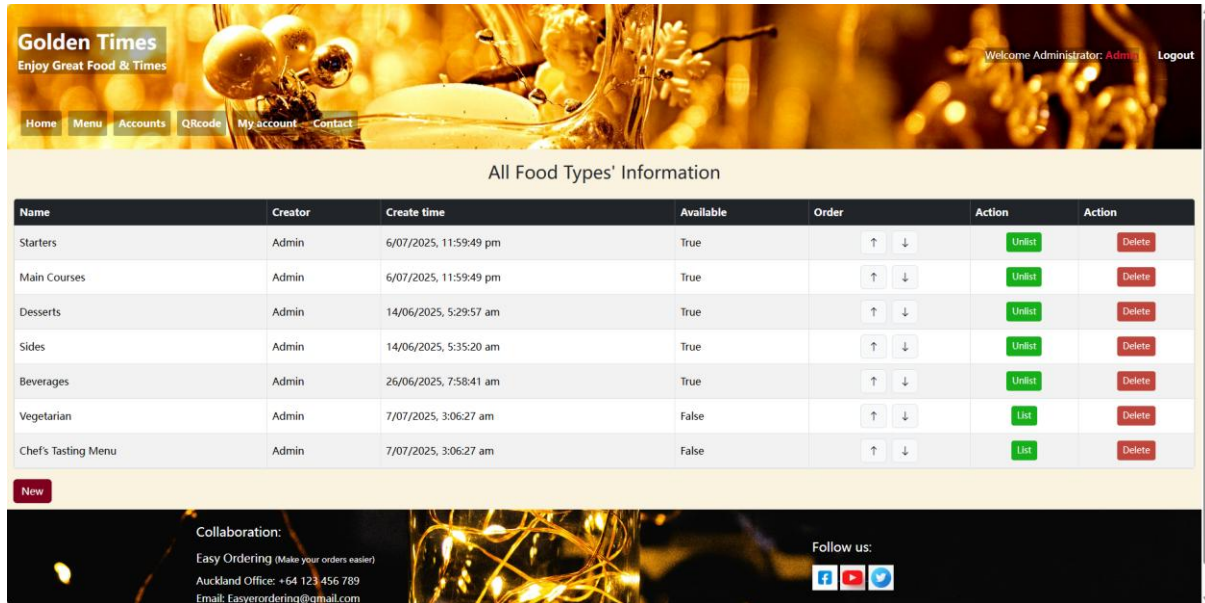
Home Menu Accounts QRcode My account Contact

Today's menu

Starters

Image	Item	Price	Discount
	Beef Carpaccio Thinly sliced raw beef served with arugula, Parmesan, and truffle oil. A refined and elegant starter.	\$12.50	50% OFF
	Smoked Salmon Cold-smoked Atlantic salmon, crème fraîche, pickled shallots, dill, with dark rye toast.	\$21.50	
	Baked Brie with Honeyed Figs Creamy French brie oven-baked to molten perfection, crowned with caramelized figs in spiced honey and crushed walnuts. Served with artisan sourdough crisps.	\$18.80	

The administrators can see all food types by clicking Menu-Food Type. They will see all the listed food types and related information. They can delete, list or unlist a food type, and adjust the order to show on the Menu page.



Golden Times
Enjoy Great Food & Times

Welcome Administrator: Admin Logout

Home Menu Accounts QRcode My account Contact

All Food Types' Information

Name	Creator	Create time	Available	Order	Action	Action
Starters	Admin	6/07/2025, 11:59:49 pm	True	↑ ↓	Unlist	Delete
Main Courses	Admin	6/07/2025, 11:59:49 pm	True	↑ ↓	Unlist	Delete
Desserts	Admin	14/06/2025, 5:29:57 am	True	↑ ↓	Unlist	Delete
Sides	Admin	14/06/2025, 5:35:20 am	True	↑ ↓	Unlist	Delete
Beverages	Admin	26/06/2025, 7:58:41 am	True	↑ ↓	Unlist	Delete
Vegetarian	Admin	7/07/2025, 3:06:27 am	False	↑ ↓	List	Delete
Chef's Tasting Menu	Admin	7/07/2025, 3:06:27 am	False	↑ ↓	List	Delete

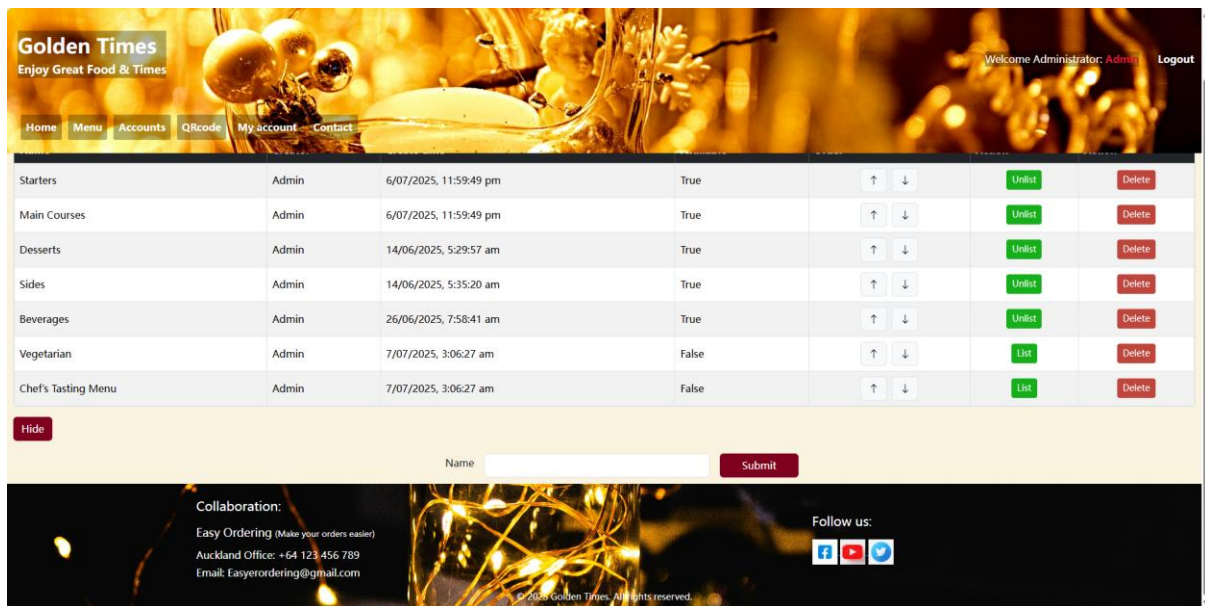
[New](#)

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Easy Ordering (Make your orders easier)
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Email: Easierordering@gmail.com

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They can create a new food type by clicking the New button, then inputting its name and clicking the Submit button.



Golden Times
Enjoy Great Food & Times

Welcome Administrator: Admin Logout

Home Menu Accounts QRcode My account Contact

All Food Types' Information

Name	Creator	Create time	Available	Order	Action	Action
Starters	Admin	6/07/2025, 11:59:49 pm	True	↑ ↓	Unlist	Delete
Main Courses	Admin	6/07/2025, 11:59:49 pm	True	↑ ↓	Unlist	Delete
Desserts	Admin	14/06/2025, 5:29:57 am	True	↑ ↓	Unlist	Delete
Sides	Admin	14/06/2025, 5:35:20 am	True	↑ ↓	Unlist	Delete
Beverages	Admin	26/06/2025, 7:58:41 am	True	↑ ↓	Unlist	Delete
Vegetarian	Admin	7/07/2025, 3:06:27 am	False	↑ ↓	List	Delete
Chef's Tasting Menu	Admin	7/07/2025, 3:06:27 am	False	↑ ↓	List	Delete

[Hide](#)

Name [Submit](#)

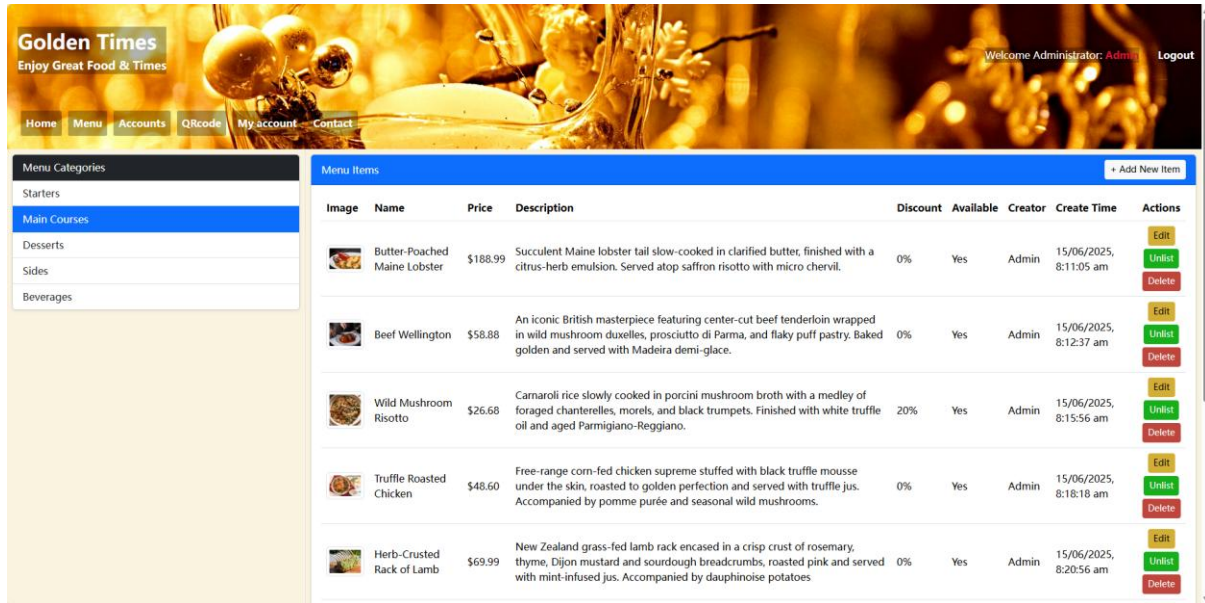
Collaboration:
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Email: Easierordering@gmail.com

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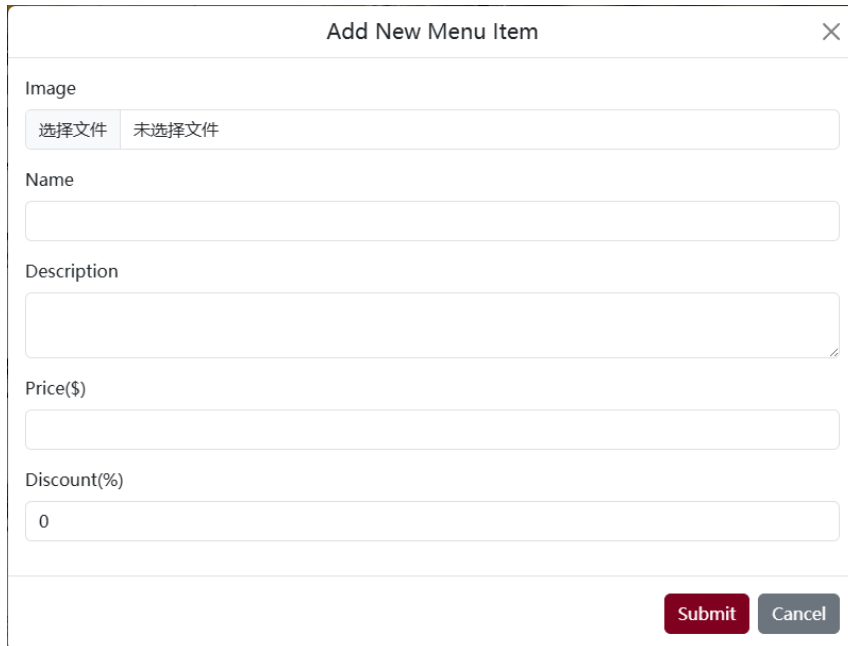
The administrators can see all items by clicking Menu-Menu. They will see all food types and related items. When they click a food type on the left, all related items will display on the right list. They can delete, list or unlist an item. They can click the image to zoom in.



The screenshot shows the 'Golden Times' restaurant management interface. The header includes the logo 'Golden Times' with the tagline 'Enjoy Great Food & Times', a navigation menu (Home, Menu, Accounts, QRcode, My account, Contact), and a user greeting 'Welcome Administrator: Admin' with a 'Logout' button. The main content area is divided into two sections: 'Menu Categories' on the left and 'Menu Items' on the right. The 'Menu Categories' section lists 'Starters', 'Main Courses' (selected), 'Desserts', 'Sides', and 'Beverages'. The 'Menu Items' section displays a table of menu items with columns for Image, Name, Price, Description, Discount, Available, Creator, Create Time, and Actions. The table contains five items: Butter-Poached Maine Lobster, Beef Wellington, Wild Mushroom Risotto, Truffle Roasted Chicken, and Herb-Crusted Rack of Lamb. Each item has an 'Edit', 'Unlist', and 'Delete' button in the Actions column.

Image	Name	Price	Description	Discount	Available	Creator	Create Time	Actions
	Butter-Poached Maine Lobster	\$188.99	Succulent Maine lobster tail slow-cooked in clarified butter, finished with a citrus-herb emulsion. Served atop saffron risotto with micro chervil.	0%	Yes	Admin	15/06/2025, 8:11:05 am	Edit Unlist Delete
	Beef Wellington	\$58.88	An iconic British masterpiece featuring center-cut beef tenderloin wrapped in wild mushroom duxelles, prosciutto di Parma, and flaky puff pastry. Baked golden and served with Madeira demi-glace.	0%	Yes	Admin	15/06/2025, 8:12:37 am	Edit Unlist Delete
	Wild Mushroom Risotto	\$26.68	Carnaroli rice slowly cooked in porcini mushroom broth with a medley of foraged chanterelles, morels, and black trumpets. Finished with white truffle oil and aged Parmigiano-Reggiano.	20%	Yes	Admin	15/06/2025, 8:15:56 am	Edit Unlist Delete
	Truffle Roasted Chicken	\$48.60	Free-range corn-fed chicken supreme stuffed with black truffle mousse under the skin, roasted to golden perfection and served with truffle jus. Accompanied by pomme purée and seasonal wild mushrooms.	0%	Yes	Admin	15/06/2025, 8:18:18 am	Edit Unlist Delete
	Herb-Crusted Rack of Lamb	\$69.99	New Zealand grass-fed lamb rack encased in a crisp crust of rosemary, thyme, Dijon mustard and sourdough breadcrumbs, roasted pink and served with mint-infused jus. Accompanied by dauphinoise potatoes	0%	Yes	Admin	15/06/2025, 8:20:56 am	Edit Unlist Delete

If they want to add an item, they can click the Add New Item button. A new dialogue will show. They can fill in all the input boxes and then click the Submit button. There are also some error checks.



The screenshot shows the 'Add New Menu Item' dialog form. It has a title bar with the text 'Add New Menu Item' and a close button (X). The form contains several input fields: 'Image' with a file selection button '选择文件' and a status '未选择文件'; 'Name' with a text input field; 'Description' with a text area; 'Price(\$)' with a text input field; 'Discount(%)' with a text input field containing the value '0'. At the bottom right, there are two buttons: 'Submit' and 'Cancel'.

If they want to edit an item, they can click the Edit button. A new dialogue will show. They can change all the input boxes and then click the Submit button. There are also some error checks.

Edit New Menu Item

Image



选择文件

未选择文件

Name

Beef Carpaccio

Description

Thinly sliced raw beef served with arugula, Parmesan, and truffle oil. A refined and elegant starter.

Food type

Starters

Price(\$)

12.50

Available

True

Discount(%)

50

Submit

Cancel

The administrators can manage QR codes by clicking QRcode. They will see all the tables and QR codes. They can delete a QR code or click the image to zoom in.

Golden Times

Enjoy Great Food & Times

Home

Menu

Accounts

QRcode

My account

Contact

Welcome Administrator: Admin

Logout

All QR Codes

Table: 1



Print

Delete

Table: 2



Print

Delete

Table: 3



Print

Delete

Table: 4



Print

Delete

Table: 5



Print

Delete

Table: 6



Table: 7



Table: 8



They can print a QR code to paste on the table.

打印

总数: 1 纸张

打印机

HP99543A (HP Smart Tank ...)

份数

1

布局

☒ 纵向

☐ 横向

页码

☒ 全部

☐ 例如1-5, 8, 11-13

颜色

颜色

双面打印

单面打印

更多设置

使用系统对话框打印... (Ctrl+Shift+P)

打印

取消

Scan to order



Table: 8

They can create a QR code by inputting a table number.

Golden Times

Enjoy Great Food & Times

Welcome Administrator: Admin Logout

Home

Menu

Accounts

QRcode

My account

Contact



Print Delete



Print Delete



Print Delete

+ Add A QR Code

Please input a table number

Table number

Generate

Collaboration:

Easy Ordering (Make your orders easier)

Auckland Office: +64 123 456 789

Email: Easierordering@gmail.com

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Customers can scan a QR code to navigate to the Menu page. The system will create a temporary name for the customers on the same table. On the page, they can change the Number and click the Add button to add items to the Cart. The total number and amount will show on the top right page dynamically. A small red total number also shows on the top right of the cart image.

Golden Times

Enjoy Great Food & Times

Welcome Customer: 17518420534646

Logout

Home

Menu

Contact

0 items

Total: \$ 0.00

W

Today's menu

Starters

Image	Item	Price	Discount	Number	Action
	Beef Carpaccio Thinly sliced raw beef served with arugula, Parmesan, and truffle oil. A refined and elegant starter.	\$12.50	50% OFF	1	Add
	Smoked Salmon Cold-smoked Atlantic salmon, crème fraîche, pickled shallots, dill, with dark rye toast.	\$21.50		1	Add
	Baked Brie with Honeyed Figs Creamy French brie oven-baked to molten perfection, crowned with caramelized figs in spiced honey and crushed walnuts. Served with artisan sourdough crisps.	\$18.80		1	Add

The customers can go to the Cart page by clicking the Cart picture. On the page, they can see all the items they selected or delete them. They can place an order after finishing the order.

Golden Times

Enjoy Great Food & Times

Welcome Customer: 17518420534646

Logout

Home

Menu

Contact

8 items

Total: \$ 707.25

W

Table 6 's Shopping Cart

Item	Price	Discount	CurPrice	Number	Status	Action
Beef Carpaccio	\$12.50	50% OFF	6.25	1	pending	Delete
Baked Brie with Honeyed Figs	\$18.80		18.80	1	pending	Delete
Butter-Poached Maine Lobster	\$188.99		188.99	1	pending	Delete
Herb-Crusted Rack of Lamb	\$69.99		69.99	1	pending	Delete
Chocolate Lava Cake	\$19.99	20% OFF	15.99	1	pending	Delete
Crispy Brussels Sprouts	\$12.50	20% OFF	10.00	1	pending	Delete
Cappuccino	\$6.80	10% OFF	6.12	1	pending	Delete
Barolo D'OCG	\$488.88	20% OFF	391.10	1	pending	Delete

Place an order

Pay

Collaboration:

Easy Ordering (Make your orders easier)

Auckland Office: +64 123 456 789

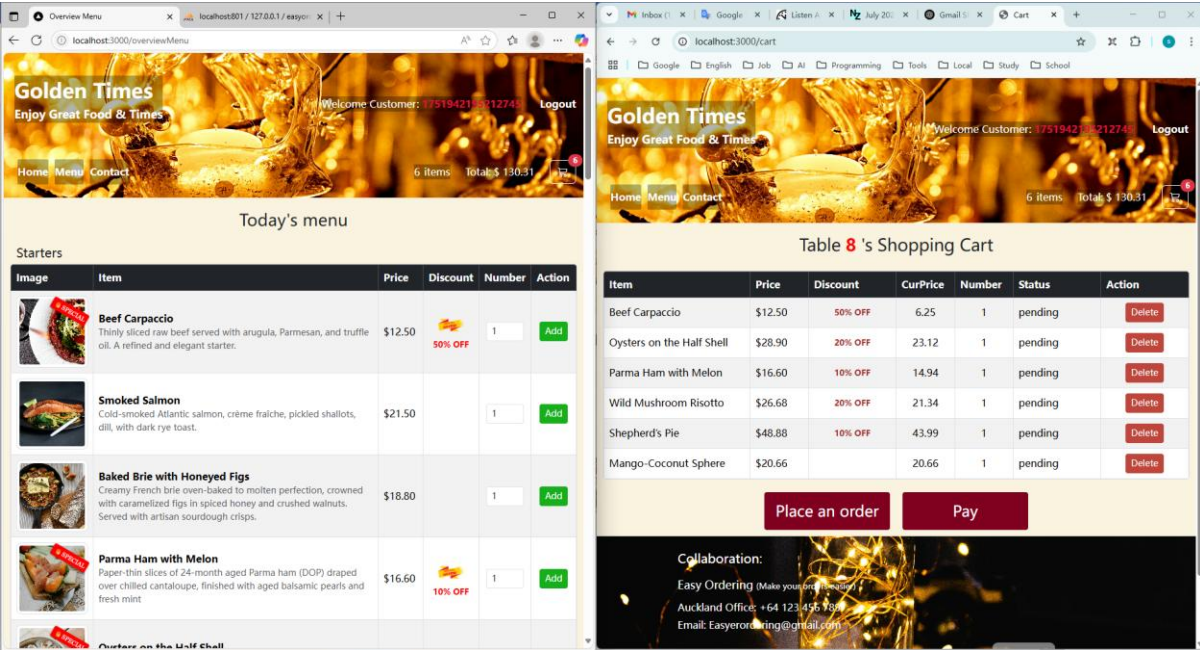
Follow us:

f

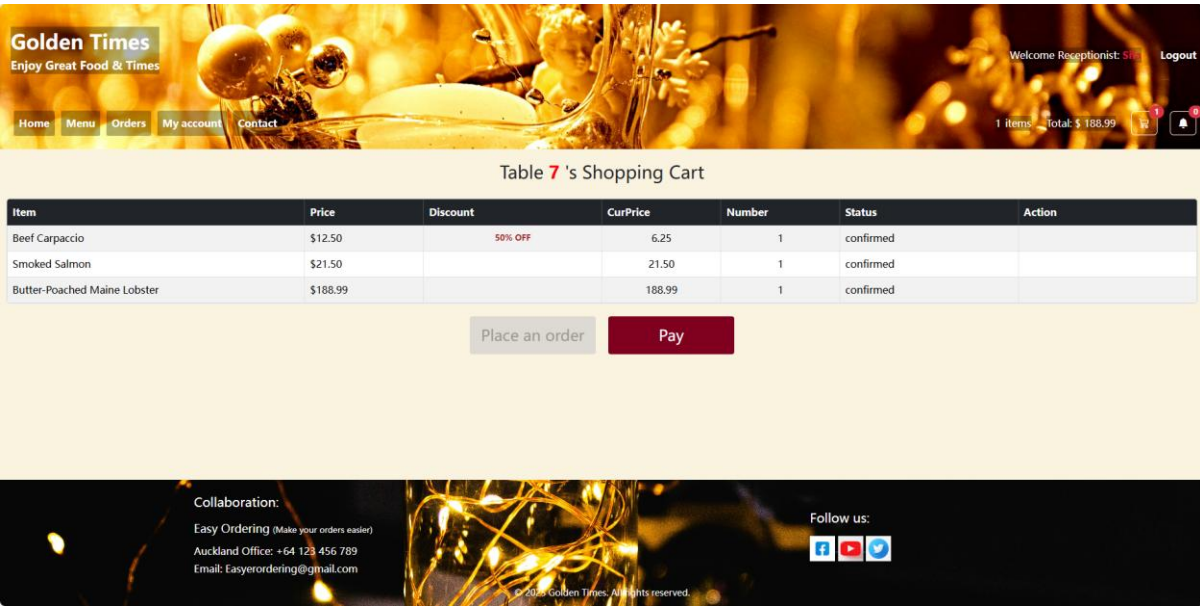
y

t

If more than one customer is in the table, they can all scan the QR code to share the same order. The following image shows two different browsers scanning Table 8's QR code. They have the same customer's name. They can all select their items and see other people's items.



After placing an order, the Place an order button will be disabled. The Delete buttons will disappear. The status of all items will change to confirmed.



They can click the Pay button to go to the Payment page. This page will show the total amount. After filling in information about their cards, they can click the Pay Now button to pay. The page will tip the result. If they pay successfully, the status of the order will change to completed.

Golden Times
Enjoy Great Food & Times

Welcome Customer: 1751804905346407 Logout

Home Menu Contact

0 items Total: \$ 707.25

Secure Payment

Card Number
1234 5678 9012 3456

Expiry MM/YY CVV 123

Amount (NZD)
707.25

Pay Now

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Email: Easierordering@gmail.com

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If the customers do not like to scan the QR code to order, they can go to the counter and ask the receptionist to make an order for them. The receptionist logs in and goes to the Menu page, where there is a table input box at the top. After they fill in the table number, they can add items to the Cart. The following operation is similar to Customers. When a receptionist logs in, there is a bell image that is shown on the right of the Cart. When customers place their orders, the number of new orders will show on the top right of the Bell image. At the same time, the system will play a voice to notify the new orders for receptionists.

Golden Times
Enjoy Great Food & Times

Welcome Receptionist: 585 Logout

Home Menu Orders My account Contact

0 items Total: \$ 0.00

Today's menu

Table:

Starters

Image	Item	Price	Discount	Number	Action
	Beef Carpaccio Thinly sliced raw beef served with arugula, Parmesan, and truffle oil. A refined and elegant starter.	\$12.50	50% OFF	1	Add
	Smoked Salmon Cold-smoked Atlantic salmon, crème fraîche, pickled shallots, dill, with dark rye toast.	\$21.50		1	Add
	Baked Brie with Honeyed Figs				Add

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The receptionists can see Today's orders by clicking Orders-Today orders. On this page, they can see all today's orders. The uncompleted orders will show on the front of the page and in red colour. They can click the Receive button to accept an order and click the Print New button to print new items for the chef.

Golden Times

Enjoy Great Food & Times

Home

Menu

Orders

My account

Contact

Welcome Receptionist: SAR

Logout

0 Items

Total: \$ 0.00

0

1

Today's orders

Table Number: 2

Status: confirmed

Number: 4

TotalAmount: \$65.81

Creator: 1751873319369869

StartTime: 2025-07-07 19:28:44

FinishTime: Invalid Date

Item	Price	Discount	CurPrice	Number	Status	Action
Beef Carpaccio	\$12.50	50% OFF	6.25	1	confirmed	Delete
Smoked Salmon	\$21.50	None	21.50	1	confirmed	Delete
Parma Ham with Melon	\$16.60	10% OFF	14.94	1	confirmed	Delete
Oysters on the Half Shell	\$28.90	20% OFF	23.12	1	confirmed	Delete

Receive

Print New

Cancel

Delete

Table Number: 5

Status: completed

Number: 6

TotalAmount: \$278.99

Creator: 1751839795318865

StartTime: 2025-07-07 10:09:57

FinishTime: 2025-07-07 11:51:06

Item	Price	Discount	CurPrice	Number	Status	Action
Butter-Poached Maine Lobster	\$188.99	None	188.99	1	completed	
Roast Beef	\$50.00	None	50.00	1	completed	

After receiving an order, the status of all items will change to preparing. The Receive button will disappear and the Ready buttons will show for every item. The receptionist can click the Ready button when the item has been served. After an item is served, the Delete button will disappear and the status of the item will change to ready.

Golden Times

Enjoy Great Food & Times

Home

Menu

Orders

My account

Contact

Welcome Receptionist: SAR

Logout

0 Items

Total: \$ 0.00

0

1

Today's orders

Table Number: 2

Status: preparing

Number: 4

TotalAmount: \$65.81

Creator: 1751873319369869

StartTime: 2025-07-07 19:28:44

FinishTime: Invalid Date

Item	Price	Discount	CurPrice	Number	Status	Action
Beef Carpaccio	\$12.50	50% OFF	6.25	1	preparing	Ready Delete
Smoked Salmon	\$21.50	None	21.50	1	preparing	Ready Delete
Parma Ham with Melon	\$16.60	10% OFF	14.94	1	preparing	Ready Delete
Oysters on the Half Shell	\$28.90	20% OFF	23.12	1	preparing	Ready Delete

Cancel

Delete

Table Number: 5

Status: completed

Number: 6

TotalAmount: \$278.99

Creator: 1751839795318865

StartTime: 2025-07-07 10:09:57

FinishTime: 2025-07-07 11:51:06

Item	Price	Discount	CurPrice	Number	Status	Action
Butter-Poached Maine Lobster	\$188.99	None	188.99	1	completed	
Roast Beef	\$50.00	None	50.00	1	completed	

The receptionist can click the Print New button to print the unreceived items for the chef. The following picture shows an order that has 5 items. Only 3 of them are new items, so just print the 3 items. The Status, the Number, the Total Amount and the FinishTime are also not printed.

Golden Times

Enjoy Great Food & Times

Home

Menu

Orders

My account

Contact

Welcome Receptionist: Sifa

Logout

3 items

Total: \$ 40.92

3

0

Today's orders

Table Number: 2

Status: preparing

Number: 5

TotalAmount: \$68.67

Creator: Sifa

StartTime: 2025-07-08 16:16:16

FinishTime:

Item	Price	Discount	CurPrice	Number	Total	Status	Action
Beef Carpaccio	\$12.50	50% OFF	\$6.25	1	\$6.25	ready	
Smoked Salmon	\$21.50	None	\$21.50	1	\$21.50	ready	
Oysters on the Half Shell	\$28.90	20% OFF	\$23.12	1	\$23.12	preparing	Ready Delete
Crispy Brussels Sprouts	\$12.50	20% OFF	\$10.00	1	\$10.00	preparing	Ready Delete
French Lentils	\$7.80	None	\$7.80	1	\$7.80	preparing	Ready Delete

Print New

Cancel

Delete

打印

总数: 1 纸张

打印机

HP99543A (HP Smart Tank ...)

份数

1

布局

纵向

横向

页面

全部

例如1-5、8、11-13

颜色

颜色

双面打印

单面打印

更多设置

使用系统对话框打印... (Ctrl+Shift+P)

打印

取消

Table Number: 2

Creator: Sifa

StartTime: 2025-07-08 16:16:16

Item	Price	Discount	CurPrice	Number	Total
Oysters on the Half Shell	\$28.90	20% OFF	\$23.12	1	\$23.12
Crispy Brussels Sprouts	\$12.50	20% OFF	\$10.00	1	\$10.00
French Lentils	\$7.80	None	\$7.80	1	\$7.80

After all items have been served, the Print New, Cancel and Delete buttons will disappear, and a Finish button will appear. The status of the order will change to ready.

Golden Times

Enjoy Great Food & Times

Home

Menu

Orders

My account

Contact

Welcome Receptionist: Sifa

Logout

0 items

Total: \$ 0.00

Today's orders

Table Number: 2

Status: ready

Number: 4

TotalAmount: \$65.81

Creator: 1751873319369869

StartTime: 2025-07-07 19:28:44

FinishTime: Invalid Date

Item	Price	Discount	CurPrice	Number	Status	Action
Beef Carpaccio	\$12.50	50% OFF	6.25	1	ready	
Smoked Salmon	\$21.50	None	21.50	1	ready	
Parma Ham with Melon	\$16.60	10% OFF	14.94	1	ready	
Oysters on the Half Shell	\$28.90	20% OFF	23.12	1	ready	

Finish

Table Number: 5

Status: completed

Number: 6

TotalAmount: \$278.99

Creator: 1751839795318865

StartTime: 2025-07-07 10:09:57

FinishTime: 2025-07-07 11:51:06

Item	Price	Discount	CurPrice	Number	Status	Action
Butter-Poached Maine Lobster	\$188.99	None	188.99	1	completed	
Beef Wellington	\$58.88	None	58.88	1	completed	

After the bill has been paid, the receptionist clicks the Finish button, and the button will disappear. The status of the order will change to completed. A Print Receipt button will appear.

Golden Times

Enjoy Great Food & Times

Home

Menu

Orders

My account

Contact

Welcome Receptionist: Sifa

Logout

0 items

Total: \$ 0.00

Table Number: 6

Status: completed

Number: 10

TotalAmount: \$1047.91

Creator: Sifa

StartTime: 2025-07-08 15:56:39

FinishTime: 2025-07-08 15:57:38

Item	Price	Discount	CurPrice	Number	Total	Status	Action
Beef Carpaccio	\$12.50	50% OFF	\$6.25	1	\$6.25	completed	
Parma Ham with Melon	\$16.60	10% OFF	\$14.94	1	\$14.94	completed	
Wild Mushroom Risotto	\$26.68	20% OFF	\$21.34	1	\$21.34	completed	
Shepherd's Pie	\$48.88	10% OFF	\$43.99	1	\$43.99	completed	
Mango-Coconut Sphere	\$20.66	None	\$20.66	1	\$20.66	completed	
Sticky Toffee Pudding	\$18.88	10% OFF	\$16.99	1	\$16.99	completed	
Dauphinoise Potatoes	\$9.80	10% OFF	\$8.82	1	\$8.82	completed	
The Perfect Martini	\$8.80	None	\$8.80	1	\$8.80	completed	
Cappuccino	\$6.80	10% OFF	\$6.12	1	\$6.12	completed	
Château Lafite Rothschild	\$999.99	10% OFF	\$899.99	1	\$899.99	completed	

Print Receipt

Collaboration:

Easy Ordering (Make your orders easier)

WhatsApp: 001 314 215 155 700

Follow us:

If the receptionist clicks the Print Receipt button, the system will print a receipt.

打印

总数: 1 纸张

打印机

HP99543A (HP Smart Tank ...)

份数

1

布局

纵向

横向

页码

全部

例如1-5, 8, 11-13

颜色

颜色

双面打印

单面打印

更多设置

使用系统默认打印机... (Ctrl+Shift+P)

打印

取消

Table Number: 6

Status: completed

Number: 10

TotalAmount: \$1047.91

Creator: Sifa

StartTime: 2025-07-08 15:56:39

FinishTime: 2025-07-08 15:57:38

Item	Price	Discount	CurPrice	Number	Total
Beef Carpaccio	\$12.50	50% OFF	\$6.25	1	\$6.25
Parma Ham with Melon	\$16.60	10% OFF	\$14.94	1	\$14.94
Wild Mushroom Risotto	\$26.68	20% OFF	\$21.34	1	\$21.34
Shepherd's Pie	\$48.88	10% OFF	\$43.99	1	\$43.99
Mango-Coconut Sphere	\$20.66	None	\$20.66	1	\$20.66
Sticky Toffee Pudding	\$18.88	10% OFF	\$16.99	1	\$16.99
Dauphinoise Potatoes	\$9.80	10% OFF	\$8.82	1	\$8.82
The Perfect Martini	\$8.80	None	\$8.80	1	\$8.80
Cappuccino	\$6.80	10% OFF	\$6.12	1	\$6.12
Château Lafite Rothschild	\$999.99	10% OFF	\$899.99	1	\$899.99

The receptionists can see History's orders by clicking Orders-History orders. After they fill in the From and To dates, they click the Search button. The system will show all orders in the special dates.

Golden Times

Enjoy Great Food & Times

Home Menu Orders My account Contact

Welcome Receptionist: Sifa Logout

0 items Total: \$ 0.00

History orders

From: yyyy/mm/日

To: yyyy/mm/日

Search

Table Number: 5

Status: completed

Number: 6

TotalAmount: \$278.99

Creator: 1751839795318865

StartTime: 2025-07-07 10:09:57

FinishTime: 2025-07-07 11:51:06

Item	Price	Discount	CurPrice	Number	Total	Status
Butter-Poached Maine Lobster	\$188.99	None	188.99	1	\$188.99	completed
Beef Wellington	\$58.88	None	58.88	1	\$58.88	completed
Truffle Fries	\$6.90	None	6.90	1	\$6.90	completed
Dauphinoise Potatoes	\$9.80	10% OFF	8.82	1	\$8.82	completed
The Perfect Martini	\$8.80	None	8.80	1	\$8.80	completed
Espresso	\$6.60	None	6.60	1	\$6.60	completed

Table Number: 6

Status: completed

Number: 100

TotalAmount: \$2312

Creator: 1751845878050538

StartTime: 2025-07-07 11:51:51

FinishTime: 2025-07-07 11:52:05

Item	Price	Discount	CurPrice	Number	Total	Status
------	-------	----------	----------	--------	-------	--------

The customers and guests can visit the Reservation page. Before reservation, they should select a date and click the Check Availability button.

Golden Times

Enjoy Great Food & Times

Home Menu Reservation Contact

Login

Table Reservation

Select Date: 2025/07/10

Check Availability

Then the page will show the availability for all tables. They can choose a table and a timeslot to book after filling in the Name, phone and Email input boxes. There is an error message if they do not fill in these boxes.

Golden Times

Enjoy Great Food & Times

Home

Menu

Reservation

Contact

Login

Table Reservation

Select Date: 2025/07/10 

Check Availability

Name: Phone: Email:

Time Slot	Table 1	Table 2	Table 3	Table 4	Table 5	Table 6	Table 7	Table 8
11:00	Reserve	Reserve	Reserve	Booked	Booked	Reserve	Reserve	Reserve
11:30	Reserve	Reserve	Reserve	Booked	Booked	Reserve	Reserve	Reserve
12:00	Reserve	Reserve	Reserve	Reserve	Booked	Reserve	Reserve	Reserve
12:30	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve
13:00	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve
13:30	Reserve	Reserve	Reserve	Booked	Reserve	Reserve	Reserve	Reserve
14:00	Reserve	Reserve	Reserve	Booked	Reserve	Reserve	Reserve	Reserve
14:30	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve
15:00	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve

If the reservation is booked successfully. The customer will receive an email about the reservation.



easierordering@gmail.com

to me 

Dear Sifa,

Date: 2025-07-10

Timeslot: 12:00



You have booked 30 mintues for Table Number: 5.
Please contact us before the timeslot if you change your plan.
Thank you for choosing EasyOrdering!

Best regards,
EasyOrdering Team

The receptionist can cancel or book a reservation for customers.

Golden Times
Enjoy Great Food & Times

Welcome Receptionist: Sifa Logout

Home Menu Orders My account Reservation Contact

0 items Total: \$ 0.00

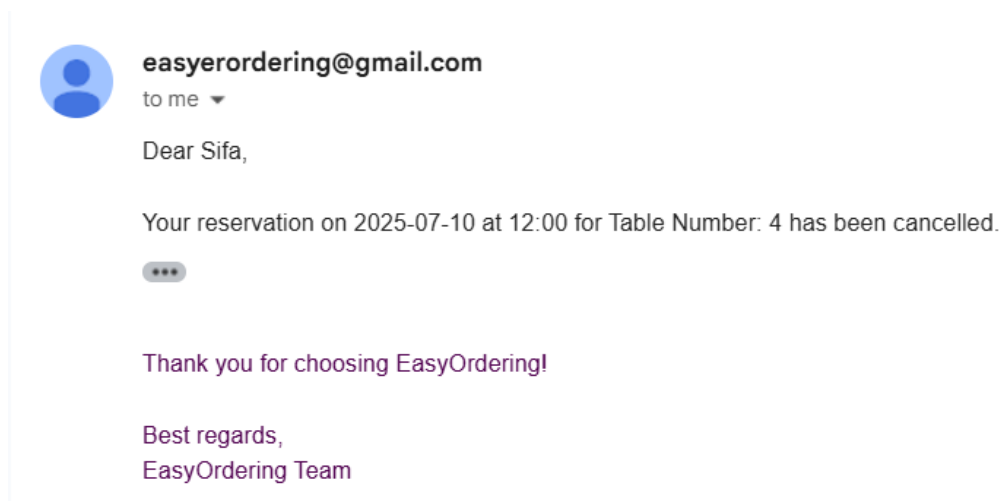
Table Reservation

Select Date: 2025/07/10 [Check Availability](#)

Name: Phone: Email:

Time Slot	Table 1	Table 2	Table 3	Table 4	Table 5	Table 6	Table 7	Table 8
11:00	Reserve	Reserve	Reserve	Cancel	Cancel	Reserve	Reserve	Reserve
11:30	Reserve	Reserve	Reserve	Cancel	Cancel	Reserve	Reserve	Reserve
12:00	Reserve	Reserve	Reserve	Reserve	Cancel	Reserve	Reserve	Reserve
12:30	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve
13:00	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve
13:30	Reserve	Reserve	Reserve	Cancel	Reserve	Reserve	Reserve	Reserve
14:00	Reserve	Reserve	Reserve	Cancel	Reserve	Reserve	Reserve	Reserve
14:30	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve
15:00	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve	Reserve

If the reservation is cancelled successfully. The customer will also receive an email.



DEVELOPER DOCUMENTATION

TOOLS AND TECHNOLOGIES USED (IN ALPHABETICAL ORDER)

- Bootstrap
- CSS
- EJS
- Express.js
- HTML
- JavaScript
- MySQL
- Node.js
- phpMyAdmin
- XAMPP
- Visual Studio Code
- Jest

OVERVIEW OF SOFTWARE AND ITS SOURCE CODE

My code is saved on GitHub. Everyone can visit it from <https://github.com/sifaZhang/EasyOrdering>. I will attach it as an appendix at the end of the document, but I would not attach the .env file because it includes my personal information. If you want to try the Reset Password function, you can write your email and password into this file.

KNOWN BUGS AND LIMITATIONS

Limitations:

1. The Reset Password feature requires an .env file that is not included in the code for security reasons. Anyone who wants to use this function must manually add their own email and app password to the file.
2. The system is designed for QR code-based ordering. Since it is not deployed online, it has not been tested on mobile devices and may display incorrectly on phones.
3. The payment feature is not functional because a payment gateway has not been integrated.

Bugs:

1. The total item count and total amount shown in the header may be incorrect if the order is modified multiple times or if the receptionist places several orders in succession.
2. If the receptionist places an order for Table 7 and then for Table 3, and later tries to add items to Table 7, the system fails to find the previous order and creates a new one instead.

SDLC

I followed the original plan outlined in my proposal. However, I made some adjustments during development when I realised certain features needed to be completed earlier. For example, I had to implement the QR code generation page before the customer ordering functionality, since customers need a QR code to scan in order to access the system.

After these adjustments, the iteration process was as follows:

Sprint 1	Header, Footer, and Navigation for different users The Login and Reset Password pages for Admin and Receptionist The accounts page is used to add or delete accounts My Account page for Admin and Receptionist to update account information
Sprint 2	Food type page for Admin to add, list, unlist, delete, and change the show orders Dishes page for Admin to add, list, unlist, delete, and edit items Menu page for Guest and Admin
Sprint 3	QR code page for Admin to add, print and delete a code Scan a code to visit the menu and add items The Cart page for customers to place an order The Payment page for customers to pay online Counting the total number and total amount to show the results on the Header
Sprint 4	The Menu page for the receptionist The Today orders page for the receptionist to change the status of orders and items The History orders page for the receptionist to search historical data New orders notification on the Header
Sprint 5	The Contact page for all users The Home page for all users to see the top ordered items

The Agile development methodology is highly effective and well-suited for software projects. In my development process, I divided all system functions into different Sprints based on their priority. This allowed me to focus on delivering specific features within short, manageable timeframes.

At the end of each Sprint, I was able to review the progress, evaluate the results, and adjust upcoming tasks and timelines based on my actual situation, such as time constraints or technical challenges. This flexibility helped me stay on track while continuously improving the system.

One of the biggest advantages of Agile is that it allows for iterative development and quick responses to changes. Instead of waiting until the end to see the final product, I could get feedback earlier and improve the system step by step.

I believe Agile not only improves productivity but also increases the quality of the final product. Based on this positive experience, I plan to continue using the Agile methodology for future software development projects.

PROJECT TIMELINE

I did not fully follow my original project timeline because I had another class starting on July 21st, which required me to complete the system more quickly. Although I spent less time than planned, I successfully implemented all the features outlined in the proposal and even added extra functionalities during development. Special thanks to ChatGPT for answering many of my questions — it greatly improved my efficiency.

ETHICAL AND CULTURAL IMPACT

The section stays the same with my proposal. Since the system does not track or profile users in any way, this issue does not apply to my system.

TEST DOCUMENT

In the project, I did Integration Testing and parts of Unit Testing. The table shows all the Integration Testing.

Test Case Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
Check Admin and Receptionist login with valid data	1. Click Login. 2. Enter username and password 3. Click login	Username and password of the register	1. Show user role and username on the top right. 2. Navigate to the home page 3. Login change to logout	1. Show user role and username on the top right. 2. Navigate to the home page 3. Login change to logout	Pass
Check Admin and Receptionist login with invalid data	1. Click Login. 2. Enter username and password 3. Click login	Username and password of the unregistered	1. Show tip: Please input the field when not inputting a username or password. 2. Show error: Incorrect username and/or Password!	1. Show tip: Please input the field when not inputting a username or password. 2. Show error: Incorrect username and/or Password!	Pass
Check the Reset password with valid data	1. Click Reset password. 2. Enter email 3. Click Reset	Email of the register	1. Show tip: successfully! Please check your email for further instructions. 2. Receive an email to show the password.	1. Show tip: successfully! Please check your email for further instructions. 2. Receive an email to show the password.	Pass
Check the Reset password with invalid data	1. Click Reset password. 2. Enter email 3. Click Reset	Email of the unregistered	1. Show error: Email not found!	1. Show error: Email not found!	Pass
Check the Home page shows for all users	1. Click Home 2. Log in as a customer. 3. Order the food at the fifth place for 100	None	1. Show the top 5 ordered items for every food type 2. The top 5 item changes to top 1.	1. Show the top 5 ordered items for every food type 2. The top 5 item changes to top 1.	Pass

	to let it become the top 1. 4. Click pictures.		3. All pictures can be zoomed in.	3. All pictures can be zoomed in.	
Check the Contact page shows for all users	1. Click Contact 2. Scroll down	None	1. Show the right emails 2. Show the right address 3. Show the right phones	Show the right emails 2. Show the right address 3. Show the right phones	Pass
Update account information with valid data	1. Log in as Admin or receptionist 2. Click My Account. 3. Input new information	Username and password of the Admin and Receptionist	1. Tip: Update success! 2. New data has been updated.	1. Tip: Update success! 2. New data has been updated.	Pass
Update account information with invalid data	1. Log in as Admin or receptionist 2. Click My Account. 3. Input a different password in the Confirm box or empty in the other box	Username and password of Admin and Receptionist	1. Show error: Passwords do not match. 2. Show tip: Please input the field when not inputting all data.	1. Show error: Passwords do not match. 2. Show tip: Please input the field when not inputting all data.	Pass
Check the current accounts	1. Log in as Admin 2. Click Accounts	Username and password of the Admin	1. The users shown on the page are the same as the database	1. The users shown on the page are the same as the database	Pass
Add an account with valid data	1. Log in as Admin 2. Click Accounts 3. Click the Add button 4. Input data 5. Click Submit	1. Username and password of the Admin 2. New account info	1. Show input controls. 2. Add a user button changes to the Hide form button 3. After inputting data, click the Submit button, then tip: Add user success! 4. The new account shows on the list. 5. Hide Form button changes to the Add a user button	1. Show input controls. 2. Add a user button changes to the Hide form button 3. After inputting data, click the Submit button, then tip: Add user success! 4. The new account shows on the list. 5. Hide Form button changes to the Add a user button	Pass
Add an account with invalid data	1. Log in as Admin 2. Click Accounts 3. Click the Add button 4. Enter a different password 5. Enter the wrong email format 6. Keep one of them empty 7. Click Submit	1. Username and password of the Admin	1. Show input controls. 2. Add a user button changes to the Hide form button. 3. Show error: Passwords do not match. 4. Email must include @. 5. Show tip: Please input the field when not inputting all data.	1. Show input controls. 2. Add a user button changes to the Hide form button. 3. Show error: Passwords do not match. 4. Email must include @. 5. Show tip: Please input the field when not inputting all data.	Pass
Delete an account	1. Log in as Admin 2. Click Accounts 3. Click the Delete button	Username and password of the Admin	1. The account has been deleted from the page. 2. The account has been deleted from the database	1. The account has been deleted from the page. 2. The account has been deleted from the database	Pass

View menu as Admin	1. Log in as Admin 2. Click Menu 3. Click Overview. 4. Click images.	Username and password of the Admin	1. Show all the listed food types and items 2. Can not choose an item. 3. All pictures can be zoomed in.	1. Show all the listed food types and items 2. Can not choose an item. 3. All pictures can be zoomed in.	Pass
Add a food type	1. Log in as Admin 2. Click Menu 3. Click food type 4. Click the New button 5. Input a name 6. Click the Submit button	Username and password of the Admin	1. The new button changes to the Hide button. 2. The page shows Name and Submit controls. 3. Tip: Add food type successfully!. 4. New food type shows on the page and database.	1. The new button changes to the Hide button. 2. The page shows Name and Submit controls. 3. Tip: Add food type success!. 4. New food type shows on the page and database.	Pass
List a food type	1. Log in as Admin 2. Click Menu 3. Click food type 4. Click List button 5. Click Overview	Username and password of the Admin	1. List button changes to Unlist button 2. The food type shows on today's menu 3. Tip: Update Available successfully!.	1. List button changes to Unlist button 2. The food type shows on today's menu. 3. Tip: Update Available successfully!.	Pass
Unlist a food type	1. Log in as Admin 2. Click Menu 3. Click food type 4. Click Unlist button 5. Click Overview	Username and password of the Admin	1. Unlist button changes to list button 2. The food type does not show on today's menu. 3. Tip: Update Available successfully!.	1. Unlist button changes to list button 2. The food type does not show on today's menu. 3. Tip: Update Available successfully!.	Pass
Delete a food type	1. Log in as Admin 2. Click Menu 3. Click food type 4. Click Delete button 5. Click Overview	Username and password of the Admin	1. The food type has been deleted from the database and the page. 2. Tip: Delete a record successfully! 3. The food type does not show on today's menu.	1. The food type has been deleted from the database and the page. 2. Tip: Delete a record successfully! 3. The food type does not show on today's menu.	Pass
Down the order of a food type	1. Log in as Admin 2. Click Menu 3. Click food type 4. Click the Down button 5. Click Overview	Username and password of the Admin	1. The place of the food type goes down to the next row. 2. The position of the food type shows on today's menu changes to the next position.	1. The place of the food type goes down to the next row. 2. The position of the food type shows on today's menu changes to the next position.	Pass
Up the order of a food type	1. Log in as Admin 2. Click Menu 3. Click food type 4. Click the Up button 5. Click Overview	Username and password of the Admin	1. The place of the food type goes up to the next row. 2. The position of the food type shows on today's menu changes to the upper position.	1. The place of the food type goes up to the next row. 2. The position of the food type shows on today's menu changes to the upper position.	Pass
Check the items	1. Log in as Admin 2. Click Menu 3. Click Menu 4. Click each food type	Username and password of the Admin	1. Show all food types. 2. Show all items under the current food type.	1. Show all food types. 2. Show all items under the current food type.	Pass

Add an item	1. Log in as Admin 2. Click Menu 3. Click Menu 4. Click Add New Item 5. Input new item information 6. Click Submit	1. Username and password of the Admin 2. New item picture.	1. Tip me to select a picture, input Description, Name, Price, Discount. 2. The new item shows on the menu page and is saved to the database.	1. Tip me to select a picture, input Description, Name, Price, Discount. 2. The new item shows on the menu page and is saved to the database.	Pass
Cancel a new item	1. Log in as Admin 2. Click Menu 3. Click Menu 4. Click Add New Item 5. Input new item information 6. Click Cancel	Username and password of the Admin	1. The new item does not show on the menu page	1. The new item does not show on the menu page	Pass
List an item	1. Log in as Admin 2. Click Menu 3. Click Menu 4. Click List 5. Go to the Overview page	Username and password of the Admin	1. Tip: Are you sure you want to list this item? 2. List changes to Unlist. 3. The item shows on Today's menu	1. Tip: Are you sure you want to list this item? 2. List changes to Unlist. 3. The item shows on Today's menu	Pass
Unlist an item	1. Log in as Admin 2. Click Menu 3. Click Menu 4. Click List 5. Go to the Overview page	Username and password of the Admin	1. Tip: Are you sure you want to unlist this item? 2. Unlist changes to list. 3. The item does not show on Today's menu	1. Tip: Are you sure you want to unlist this item? 2. Unlist changes to list. 3. The item does not show on Today's menu	Pass
Edit an item	1. Log in as Admin 2. Click Menu 3. Click Menu 4. Click Edit 5. Input new data and click Submit. 6. Go to the Overview page	Username and password of the Admin	1. Show Edit New Menu Item dialogue with current item data. 2. New data shows on this page, Overview page and the database	1. Show Edit New Menu Item dialogue with current item data. 2. New data shows on this page, Overview page and the database	Pass
Cancel the editing an item	1. Log in as Admin 2. Click Menu 3. Click Menu 4. Click Edit 5. Input new data and click Cancel. 6. Go to the Overview page	Username and password of the Admin	1. Show Edit New Menu Item dialogue with current item data. 2. New data does not show on this page, the Overview page and the database	1. Show Edit New Menu Item dialogue with current item data. 2. New data does not show on this page, the Overview page and the database	Pass
Delete an item	1. Log in as Admin 2. Click Menu 3. Click Menu 4. Click Delete 5. Go to the Overview page	Username and password of the Admin	1. Tip: Are you sure you want to delete this item? 2. The item does not show on this page and Today's menu	1. Tip: Are you sure you want to delete this item? 2. The item does not show on this page and Today's menu	Pass
Check the special tip on items	1. Log in as Admin 2. Click Menu 3. Click Menu	1. Username and password of the Admin. 2. Some items have discounts.	1. A special picture is shown on the top right of its image. 2. Another special picture is shown in the Discount column.	1. A special picture is shown on the top right of its image. 2. Another special picture is shown in the Discount column.	Pass

			3. The words show in Discount column is red.	3. The words show in Discount column is red.	
Show all QR codes	1. Log in as Admin 2. Click QRCode. 3. Click pictures.	Username and password of the Admin	1. Show all QR codes from the database. 2. All pictures can be zoomed in.	1. Show all QR codes from the database. 2. All pictures can be zoomed in.	Pass
Create a QR code	1. Log in as Admin 2. Click QRCode 3. Input a table number 4. Click Generate	Username and password of the Admin	1. Generate a new QR code on this page. 2. The new code is saved on the server. 2. Scan the code link to the app.	1. Generate a new QR code on this page. 2. The new code is saved on the server. 2. Scan the code link to the app.	Pass
Delete a QR code	1. Log in as Admin 2. Click QRCode 3. Click Delete	Username and password of the Admin	1. The code is deleted from this page and the database.	1. The code is deleted from this page and the database.	Pass
Print a QR code	1. Log in as Admin 2. Click QRCode 3. Click Print	Username and password of the Admin	1. Show the Print dialogue. 2. Print the QR code when I click the Print button.	1. Show the Print dialogue. 2. Print the QR code when I click the Print button.	Pass
View the menu as a Guest	1. Click Menu. 2. Click images.	None	1. Show all the listed food types and items 2. Can not choose an item. 3. All pictures can be zoomed in.	1. Show all the listed food types and items 2. Can not choose an item. 3. All pictures can be zoomed in.	Pass
View the menu as a Customer	1. Scan a QR code. 2. Click images.	None	1. Show all the listed food types and items 2. There are a Number input box and an Add button for each item. 3. All pictures can be zoomed in.	1. Show all the listed food types and items 2. There are a Number input box and an Add button for each item. 3. All pictures can be zoomed in.	Pass
Add items to the cart as a customer	1. Scan a QR code. 2. Click the Add buttons. 3. Click the Cart button	None	1. Show the total number and total amount on the top right of the page dynamically when adding items. 2. New items show on the cart page.	1. Show the total number and total amount on the top right of the page dynamically when adding items. 2. New items show on the cart page.	Pass
View the cart as a Customer or Receptionist	1. Click the cart picture.	Username and password of the Receptionist	1. Show all items and their statuses.	1. Show all items and their statuses.	Pass
Place an order as a Customer or Receptionist	1. Click the cart picture. 2. Click the Place an order button.	Username and password of the Receptionist	1. Tip: Are you sure you want to place the order? 2. The statuses of all items under pending change to confirmed. 3. The Place an order button is disabled. 4. The Delete buttons disappear.	1. Tip: Are you sure you want to place the order? 2. The statuses of all items under pending change to confirmed. 3. The Place an order button is disabled. 4. The Delete buttons disappear.	Pass
Delete an item as a Customer or Receptionist	1. Click the cart picture.	Username and password of	1. Tip: Are you sure you want to delete this item?	1. Tip: Are you sure you want to delete this item?	Pass

	2. Click a Delete button.	the Receptionist	2. The item is deleted from this page and the database. 3. The total number and amount update.	2. The item is deleted from this page and the database. 3. The total number and amount update.	
Append items as a Customer or Receptionist	1. Click Menu. 2. Add items. 3. Click Cart.	1. Username and password of the Receptionist. 2. An existing order that has been confirmed or is ready	1. New items are appended to the end of the list. 2. Place an order can press. 3. Total number and amount just statistic the new items	1. New items are appended to the end of the list. 2. Place an order can press. 3. Total number and amount just statistic the new items	Pass
Delete all appended items as a Customer or Receptionist	1. Click the cart picture. 2. Click the Delete button on pending items.	1. Username and password of the Receptionist. 2. An existing order that has been confirmed or ready	1. Tip: Are you sure you want to delete this item? 2. The item is deleted from this page and the database. 3. The total number and amount update. 4. The place an order button is disabled.	1. Tip: Are you sure you want to delete this item?--pass 2. The item is deleted from this page and the database.--pass 3. The total number and amount update.--pass 4. The place an order button is disabled.--fail	Fail
More than one person selects items at the same time	1. Person A Scans a QR code. 2. Person A clicks the Add button. 3. Person B scans a QR code. 4. Person B clicks the Add button. 5. Click the Cart button	None	1. All people can see what items have been chosen by other people. 2. Each person can add or delete items 3. Each person can place an order or pay online.	1. All people can see what items have been chosen by other people. 2. Each person can add or delete items 3. Each person can place an order or pay online.	Pass
Pay online	1. Click the Cart. 2. Click the Pay button. 3. Input the card information. 4. Click the Pay Now button. 5. Click Cart.	1. An existing order that has been confirmed or ready. 2. Credit card	1. Show Payment page. 2. Show the correct money. 3. Tip: Confirm payment? 4. Tip: Pay success! 5. Nothing shows on Cart page.	1. Show Payment page. 2. Show the correct money. 3. Tip: Confirm payment? 4. Tip: Pay success! 5. Nothing shows on Cart page.	Pass
Clear pictures on the timer	1. Delete an item. 2. Delete a QR code. 3. Change the current time to 01:59 AM	1. Existing items. 2. Existing QR codes	1. All images related to the items or QR codes are deleted at 02:00.	1. All images related to the items or QR codes are deleted at 02:00.	Pass
Choose items for customers	1. Log in as Receptionist. 2. Click Menu. 3. Input the table number. 4. Click Add Items. 5. Click the Cart button	Username and password of the Receptionist	1. Show the total number and total amount on the top right of the page dynamically when adding items. 2. New items show on the cart page.	1. Show the total number and total amount on the top right of the page dynamically when adding items. 2. New items show on the cart page.	Pass

New orders notification	1. Click the Bell image.	1. Username and password of the Receptionist. 2. Existing customers have placed orders.	1. The top right of the Bell image shows the number of new orders. 2. Notify by voice when new orders are coming. 3. Show new orders when clicking the Bell.	1. The top right of the Bell image shows the number of new orders. 2. Notify by voice when new orders are coming. 3. Show new orders when clicking the Bell.	Pass
Receive an order	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders. 4. Click Receive.	1. Username and password of the Receptionist. 2. Existing a new order	1. The Receive and Print New buttons disappear. 2. The status of the order and all items change to confirmed.	1. The Receive and Print New buttons disappear. 2. The status of the order and all items change to confirmed.	Pass
Ready an item	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders. 4. Click a Ready button. 5. Click all Ready buttons.	1. Username and password of the Receptionist. 2. Existing a new order	1. The Ready and Delete buttons disappear. 2. The status of the item changes to preparing. 3. When all items have been served, the status of the order changes to ready, and the delete and cancel buttons disappear.	1. The Ready and Delete buttons disappear. 2. The status of the item changes to preparing. 3. When all items have been served, the status of the order changes to ready, and the delete and cancel buttons disappear.	Pass
Delete an item	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders. 4. Click Delete on items.	1. Username and password of the Receptionist. 2. Existing a new order	1. Tip: Are you sure you want to delete this item? 2. The item is deleted from the database and this page.	1. Tip: Are you sure you want to delete this item? 2. The item is deleted from the database and this page.	Pass
Print new items	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders. 4. Click Print New.	1. Username and password of the Receptionist. 2. Existing a new order	1. Print all items that have not been served.	1. Print all items that have not been served.	Pass
Delete an order	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders. 4. Click Delete on orders.	1. Username and password of the Receptionist. 2. Existing a new order	1. Tip: Are you sure you want to delete this order? 2. The order is deleted from the database and this page.	1. Tip: Are you sure you want to delete this order? 2. The order is deleted from the database and this page.	Pass
Cancel an order	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders. 4. Click Cancel.	1. Username and password of the Receptionist. 2. Existing an order on preparing	1. Tip: Are you sure you want to cancel this order? 2. The status of the order changes to Cancelled. 3. The status of all items that are not served changes to Cancelled.	1. Tip: Are you sure you want to cancel this order? 2. The status of the order changes to Cancelled. 3. The status of all items that are not served changes to Cancelled.	Pass
Finish an order	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders. 4. Click Finish.	1. Username and password of the Receptionist.	1. The Finish button disappears. 2. The status of the order and all items change to completed.	1. The finish button disappears. 2. The status of the order and all items change to completed.	Pass

		2. An existing order whose items have been served.			
Print a receipt	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders. 4. Click Print Receipt.	1. Username and password of the Receptionist. 2. Existing a finished order	1. Print a receipt that has all items, price, number, date, total number, and total amount.	1. Print a receipt that has all items, price, number, date, total number, and total amount.	Pass
Append items to an order under ready	1. Log in as Receptionist. 2. Click Menu. 3. Input table number. 4. Click Add items. 5. Click Cart. 6. Click Place an order.	1. Username and password of the Receptionist. 2. Existing a ready order. 3. Reception is on the Today Orders page.	1. New items show at the end of the list. 2. The status of the order changes to Confirmed. 3. Show Print New, Receive, Delete and Cancel buttons. 4. The finish button disappears.	1. New items show at the end of the list.--pass 2. The status of the order changes to Confirmed.--pass 3. Show Print New, Receive, Delete and Cancel buttons. 4. The finish button disappears.	Pass
View today's orders	1. Log in as Receptionist. 2. Click Orders. 3. Click Today Orders.	1. Username and password of the Receptionist.	1. Show all today's orders. 2. All the data is right in all orders.	1. Show all today's orders. 2. All the data is right in all orders.	Pass
View history's orders with valid dates	1. Log in as Receptionist. 2. Click Orders. 3. Click History Orders. 4. Input From date and To date. 5. Click Search.	1. Username and password of the Receptionist. 2. Existing history orders.	1. Show all orders during the period from From and To date. 2. Don't have any buttons.	1. Show all orders during the period from From and To date. 2. Don't have any buttons.	Pass
View history's orders with invalid dates	1. Log in as Receptionist. 2. Click Orders. 3. Click History Orders. 4. Click Search.	1. Username and password of the Receptionist. 2. Existing history orders.	1. Tip: Please input From and To dates	1. Tip: Please input From and To dates	Pass
Make a reservation with valid data	1. Click Reservation. 2. Fill in the date. 3. Click Check Availability.	1. A Guest or Customer or Log in as Receptionist. 2. Admin has created some tables	1. The reservation has been confirmed. 2. The customer receives an email about the reservation.	1. The reservation has been confirmed. 2. The customer receives an email about the reservation.	Pass
Make a reservation with invalid data	1. Click Reservation. 2. Click Check Availability.	1. A Guest or Customer or Log in as Receptionist. 2. Admin has created some tables	1. Tip: Please enter your name, phone, and email.	1. Tip: Please enter your name, phone, and email.	Pass
Cancel a reservation	1. Click Reservation. 2. Fill in the date.	1. Log in as Receptionist.	1. Tip: Are you sure you want to delete this reservation?	1. Tip: Are you sure you want to delete this reservation?	Pass

	3. Click Check Availability. 4. Click the Cancel button.	2. Some tables have been booked.	2. The reservation has been cancelled. 3. The customer receives an email about the cancellation.	2. The reservation has been cancelled. 3. The customer receives an email about the cancellation.	
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I use Jest to do Unit Testing. There are some screenshots.

```
tests > JS app.test.js > describe('GET /') callback > it('should render home with topNItemsPerType on success') c

1  const { toLocalDatetimeString } = require('../app');
2  const request = require('supertest');
3  const express = require('express');
4  const conn = require('../dbConfig');
5  app = require('../app');
6
7  test('should convert UTC string to local datetime string', () => {
8    const result = toLocalDatetimeString('2024-07-01 16:00:00');
9    expect(result).toMatch(/\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2}/);
10   expect(result).toBe('2024-07-02 04:00:00');
11 });
12 test('should handle invalid date string gracefully', () => {
13   const result = toLocalDatetimeString('invalid-date');
14   expect(result).toBe('Invalid Date');
15 });
16
17
18 // Mock dbConfig and its query method
19 jest.mock('../dbConfig', () => ({
20   query: jest.fn(),
21 }));
22
23
24 beforeEach(() => {
25 }
```

```
27 describe('GET /', () => {
28   afterEach(() => {
29     jest.clearAllMocks();
30   });
31
32   it('should render home with topNItemsPerType on success', async () => {
33     // Mock the db query to call the callback with no error and fake results
34     conn.query.mockImplementation((sql, params, cb) => {
35       cb(null, [{ id: 1, name: 'item1' }, { id: 2, name: 'item2' }]);
36     });
37
38     // Mock res.render
39     const res = await request(app).get('/');
40
41     expect(conn.query).toHaveBeenCalledWith(
42       'CALL SelectTopItemsByType(?)',
43       [5],
44       expect.any(Function)
45     );
46     // Since EJS rendering returns HTML, check for some expected content
47     expect(res.statusCode).toBe(200);
48     // Optionally, check that the rendered HTML contains expected data
49     // expect(res.text).toContain('item1');
50   });
51
52   it('should return 500 if database error occurs', async () => {
53     conn.query.mockImplementation((sql, params, cb) => {
54       cb(new Error('DB error'));
55     });
56
57     const res = await request(app).get('/');
58     expect(res.statusCode).toBe(500);
59     expect(res.text).toContain('Database error (SelectTopItemsByType)');
60   });
61 }
```


USER DOCUMENTATION

The system includes a default administrator account with the username Admin and password 666. When the website is first accessed, users are directed to the Home page. A Login link is displayed at the top-right corner of the page. Clicking this link will take the user to the login page, where they can log in using the default admin credentials.

Once logged in as an administrator, the user can:

- Manage food types and menu items
- Update their account information
- Create or delete other administrator or receptionist accounts
- Generate and print QR codes to place on dining tables

On the public-facing website, all users—regardless of login status—can view the Home, Contact, and Menu pages.

Customer Functionality

Customers can access the ordering system by scanning the QR code on the table. After connecting, they can:

- Browse the menu and select dishes
- View the total quantity, total amount, and status of their ordered items
- Pay the bill online after finishing their meal

Receptionist Functionality

Receptionists have additional privileges beyond regular users. On the Menu page, they can place an order by manually entering a table number (this feature is only available to receptionists). Like administrators, they can update their personal information.

Receptionists also have access to the Today's Orders page, where they can:

- Receive, cancel, delete, or complete orders
- Mark items as “ready” once they’ve been served
- Delete items that have not been served
- Print new items for the kitchen
- Print receipts for customers

Additionally, receptionists can search and view past orders on the Historical Orders page.

CONFIGURATION GUIDE

In order to send emails normally, the Administrator should have an email account for service.

1. Create a Google email account and generate an App Password.
2. Create a .env file in the project root folder.
3. Replace XXXXXXXXXXXX with your email and App Password.

```
GMAIL_APP_PASSWORD=XXXXXXXXXXXXX  
GMAIL_USER=XXXXXXXXXXXXX@gmail.com
```

FUTURE SOFTWARE ENHANCEMENT LIST

1. Add a data analytics feature to generate business insights as the customer base grows.
2. Implement a review system to gather customer feedback and improve service quality.
3. Develop an employee management module to help the restaurant manage staff information online.

REFLECTION

I began this course with only a basic understanding of web application development. A few years ago, I attempted to learn HTML, CSS, and JavaScript on my own, but the process was quite challenging. I often found myself forgetting what I had learned, and at times it felt like I had never studied them at all.

During this class, I had the opportunity to follow the Mini-Book, which gave me a structured and practical starting point for web development. Through the course, I learned how to build a full-stack application using technologies like Node.js, Bootstrap, MySQL, and Express.

In my final assignment, I was able to plan, design, develop, and test a complete web application independently. This hands-on project helped me gain a much deeper understanding of how a web system is structured and how its components work together.

I also learned how to write Use Cases and use popular project management tools such as Trello, Miro, and Jira to organise and track my progress. These tools not only improved my productivity but also gave me a clearer view of real-world software development workflows.

Although the learning curve was steep and I had to complete the project earlier due to scheduling constraints, I didn't find it too difficult. This is partly because of my previous experience in software development using C++, and also because AI tools like ChatGPT greatly supported me during the learning and debugging process.

Overall, this course significantly enhanced my practical skills and confidence in web development. It inspired me to continue learning and building more complex systems in the future.

If I am to do this project or a similar project again, there are several improvements I would make based on the experience I've gained. First, I would separate the frontend and backend into two independent layers. This would make the architecture more modular, maintainable, and scalable. It would also allow for more flexibility in choosing frontend technologies such as React or Vue.js. Second, I would optimise the code structure by organising files and modules more clearly. This would greatly improve code readability, maintainability, and help reduce technical debt. In my current project, I only had three JavaScript files, and most of the code was written directly in app.js. As the project grew larger, this made it become harder to manage and debug. In the future, I would separate different functionalities into multiple modules—for example, splitting routes, database access, and business logic into their own files—to make the structure more modular and scalable. Additionally, I would focus more on UI/UX design, including layout, responsiveness, and accessibility. A better-designed interface would enhance the user experience and make the system more professional and user-friendly. Finally, I would also consider writing more unit tests and automating part of the development and deployment process to improve system stability and efficiency.

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APPENDIX: SOURCE CODE



EasyOrdering_Final.7z